

The Margins of Firm Tax Incentives:

Behavior, avoidance, and selection of small firms *

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Abstract

Governments in developing countries extensively use preferential tax regimes to promote the registration, growth, and employment of small and medium-sized firms, with mixed results. Using novel linked microdata, we study Brazil's 2018 Factor R reform, which granted a large revenue-tax cut to firms whose payroll reached 28% of revenue. We develop a framework in which the reform's value to firms governs both incumbent behavior and selection: firms benefit either by naturally satisfying the rule through high labor intensity or by profitably manufacturing eligibility. Incumbent firms that automatically received the tax cut increased reported revenue by 10%, with no employment response. When eligibility instead required behavioral adjustment, firms overwhelmingly chose the cheapest margin, shifting owner dividends to wages rather than increasing employee payroll. Entry increased by about 50%, but selection operated through the same low-cost margin: marginal entrants were selected on income-shifting opportunities rather than labor intensity, and 35% of marginal entrants represent reallocation from another tax regime rather than additional entry. Thus, policy outcomes depend jointly on the elasticity of the real behavior policymakers seek to affect, the availability of alternative adjustment margins and the enforcement environment, and the composition of firms induced to participate.

JEL Codes: D3, H2, H3, H5

*Corresponding author: Javier Feinmann (jfeinmann@berkeley.edu). All data work for this project involving confidential taxpayer information was done at the Federal Revenue of Brazil (*Receita Federal do Brasil*, RFB) facilities, on RFB computers, by RFB tax auditors, and at no time was confidential taxpayer data ever outside of the RFB computing environment. All results have been reviewed to ensure that no confidential information is disclosed. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the *Receita Federal do Brasil*, the World Bank and its affiliated organizations, the Executive Directors of the World Bank, or the governments they represent. We thank the Stone Center for financial support.

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1 Introduction

Small and medium-sized firms (SMEs) play a central role in developing countries. Therefore, governments implement numerous policies to promote SME registration, growth, and employment. Their success depends not only on how firms respond to the incentives they create, but also on which firms they induce to participate. These two margins become especially important when preferential treatment is conditioned on criteria that firms can adjust to meet. Firms may respond along margins other than those intended, while preferential regimes may disproportionately attract firms for which those adjustments are cheapest.

On the one hand, the development literature has studied how reducing compliance costs and tax burdens can foster registration, growth, and employment among SMEs (De Andrade, Bruhn, and McKenzie, 2016; Benhassine et al., 2018; Ulyssea, 2020; Farias and Hsu Rocha, 2026). On the other hand, the public finance literature has alerted us that low monitoring and reporting environments can lead to avoidance and evasion strategies that deviate from intended policy goals (Agostini et al., 2018; Best et al., 2015; Gravouille and Stantcheva, 2025). Yet we know much less about how these forces interact: whether policies that make preferential regimes more attractive induce the desired behavioral responses, and which firms are ultimately selected into them.

This paper studies this question, bridging the two lines of research, by combining new tax microdata for Brazil with the "Factor R" reform. Starting in January 2018, the policy states that eligible firms with a *payroll-to-revenue* ratio (Factor R) of 28% or higher face a large drop in their revenue tax rate, averaging 9 pp, introducing a massive tax notch in the system. This made *Simples Nacional* (Brazil's STR) more generous through two distinct channels: firms could benefit either because they were sufficiently labor-intensive to meet the threshold, or because they could meet it at low cost by adjusting their behavior. Importantly, the payroll considered in the Factor R included both employees' payroll and owner wages, and the reform affected both incumbent and new firms.

The Factor R reform is an unprecedented tax policy that offers a unique lens for studying firms' behavioral responses to STR incentives. First, for incumbent firms, it generated two clean quasi-experiments: On the one hand, firms with pre-policy Factor R greater than 28% automatically become eligible for the revenue tax cut, thereby providing exogenous variation in the tax rate they face. On the other hand, firms with a pre-policy Factor R below 28% must increase their payroll-to-revenue ratio to receive the tax cut, allowing us to study the cheapest way firms have to reach the threshold. Second, for new firms, the policy made the STR more attractive for labor-intensive firms as well as those that can take advantage of the income-shifting loophole, allowing us to measure which group responds

the most and their corresponding counterfactual.

A common challenge in studying firms in STRs is the limited microdata available because they are typically subject to few tax forms and reporting requirements. Our analysis relies on novel tax microdata on the universe of SME firms, social security records, electronic payment systems (including invoices), bank account flows, credit card information, ownership records, and individual income tax forms.¹ This allows us to create a unique data environment with comprehensive information for the universe of firms and owners in the entire country.

We begin by showing descriptively that the large tax notch induced by the Factor R policy led to strong behavioral responses. Since the policy's implementation, we document a large bunching at 28% and excess mass just above it, and an almost fully dominated region below it. Interestingly, the cross-sectional pattern grows over time, consistent with both incumbents' learning and compositional change among new entrants, which we investigate further.

Before turning to the empirical analysis, we develop a conceptual framework that provides a common interpretation of the reform's intensive- and extensive-margin effects. The key object is the increase in the value of operating under *Simples Nacional* generated by Factor R. We decompose this value into the benefit of receiving the lower revenue-tax schedule and the optimized cost of satisfying the payroll-to-revenue eligibility requirement. This distinction highlights two ways in which firms can benefit from the reform: some naturally satisfy the rule because their production technology generates a high payroll share, while others can profitably manufacture eligibility by adjusting the Factor R components. For incumbent firms, the framework predicts how these incentives map into movements around the 28% threshold and into the margins used to reach it. For potential entrants, it governs selection into the regime, while productivity sorting helps characterize whether marginal entrants would otherwise remain unregistered or switch tax regimes.

We first turn to incumbent firms' behavioral responses, for which the reform generates two distinct quasi-experiments depending on the pre-policy location in the Factor R distribution. Firms initially below the 28% threshold can obtain the tax cut by increasing employee payroll, reducing reported revenues, shifting owner income from dividends to wages, or combining these margins. For firms with a pre-policy Factor R above the threshold, we show that the reform simply reduced the revenue tax rate, provided firms did not become constrained by the threshold, thereby helping us measure the revenue and input-distortion effects of revenue taxation.

Through a series of *difference-in-differences* analyses, we show that Low-FR firms re-

¹ Accessing this data has been possible thanks to the researcher collaboration between the International Tax Observatory (ITO) and *Receita Federal de Brasil* (RFB).

spond almost entirely through income shifting. On average, firms gradually increase owner wages, reaching 25% two years after the reform, while keeping revenues and employees' payroll and revenues constant. High-FR firms, in contrast, increase reported revenues by about 10% two years after the reform, with no detectable response in employees' payroll. Owner wages also increase, but by substantially less than for Low-FR firms; this response is concentrated among firms initially close to the threshold, consistent with the eligibility constraint becoming relevant as they re-optimize. As a result, incumbent firms' reoptimization contributes to the bunching pattern. Our decomposition analysis shows that 84% of incumbent bunchers would otherwise have been located below the threshold, while the remaining 16% would have been located above it.

We then study the extensive margin responses, as the same policy-induced increase in the value of *Simples Nacional* that governs incumbent incentives also changes firms' regime choices. In this case, we also rely on *difference-in-differences*, but we follow exposed vs. non-exposed industries, rather than firms. Additionally, We decompose the ITT estimates into the policy's effect on inframarginal entrants (always-takers) from changes in the composition of marginal entrants (compliers). This allows us to ask whether the firms induced into *Simples* are selected primarily on the labor-intensity margin targeted by the policy or on their ability to exploit the owner-income-shifting opportunity.

We find an increase in the number of new firms of about 50% in exposed industries relative to non-exposed industries. Consistent with the incentives generated by the policy, the share of new entrants with a Factor R above 28% doubles. We find that the new firms induced by the policy are three times as likely to be above 28% as the pre-reform new entrants. Decomposing Factor R reveals the source of this selection: marginal entrants differ sharply in owner wages relative to revenues, while their employees' payroll-to-revenue ratio is essentially unchanged. Thus, we find no evidence that the extensive-margin response is driven by selection of more hired-labor-intensive firms. Instead, the marginal entrants are disproportionately firms using owner compensation to satisfy Factor R, consistent with selection on the income-shifting opportunity created by the reform.

Looking at new firms' counterfactual, we find no evidence of industry, or labor contract relabeling.² However, we find a large reduction in the number of new firms in the exposed industries in *Lucro Presumido*, the main alternative tax regime. This reduction accounts for 35% of the flow of new firms in *Simples Nacional*, and a similar pattern is observed in the evolution of the stocks. Therefore, we conclude that the remaining share corresponds to true new business registration, but we are unable to determine whether it is due to formalization or business creation.

²In Brazil, a common evasion practice for employees is to arrange with their employers a B2B contract rather than an employment contract to save social security contributions and personal income taxes. This is a practice known as *pejotizacao*.

Finally, we summarize and compare the order of magnitude of each behavioral response by examining their impact on government revenues. We find that the extensive-margin responses are large, accounting for half of the mechanical effect. However, as those whose counterfactual would be the alternative tax regime contribute negatively to tax collection, the net effect is much smaller. For the incumbent firms, we show that the income-shifting response dominates the scene, as it is orders of magnitudes larger than the revenue collection for the increase in reported revenue of High FR firms.

Our results show that large tax incentives for SMEs need not translate into the intended real responses of the policy. Among incumbent firms that automatically receive the tax cut, we find a sizable increase in reported revenue but no detectable response in payroll. When obtaining the same tax cut instead requires firms to raise their payroll-to-revenue ratio, they overwhelmingly do so through income shifting rather than by hiring more labor. The same distinction appears on the extensive margin: although the reform made the regime particularly valuable to both labor-intensive firms and firms able to satisfy the eligibility rule at low cost, the firms induced to enter are selected almost entirely on the latter margin. Thus, policy outcomes depend jointly on the elasticity of the real behavior policymakers seek to affect, the availability of alternative adjustment margins, and the composition of firms induced to participate for a given enforcement environment. More broadly, large take-up of a preferential policy need not imply a correspondingly large response along the economic margin it is designed to favor.

Relation to the literature

Our paper relates first to the public-finance literature on business taxation under imperfect enforcement. A central insight of this literature is that the desirability of a tax base depends not only on the distortions it creates to real activity, but also on how easily taxpayers can manipulate it. [Best et al., 2015](#) provide an important benchmark: when profits are difficult to verify, turnover taxation can increase revenue efficiency by restricting opportunities to evade, despite creating greater production distortions than a profit tax. Related work documents that firms respond to taxation and enforcement through multiple reporting margins, including sales and cost misreporting, movements into informality, and changes in organizational form ([Carrillo, Pomeranz, and Singhal, 2017](#); [Waseem, 2018](#); [Bachas and Soto, 2021](#); [Lobel, Scot, and Zúniga, 2024](#)). Our paper studies a distinct policy problem. Simplified tax regimes do not simply replace a difficult-to-enforce profit base with an easier-to-monitor turnover base for a fixed population of firms. They offer preferential tax treatment and lower compliance requirements to a selected set of SMEs, thereby affecting both firms' behavior conditional on participation and their incentives to enter the regime in the first place. The relevant policy tradeoff therefore extends beyond the

production-versus-revenue-efficiency margin emphasized by [Best et al., 2015](#): the benefits of attracting additional firms into a simplified regime must be evaluated together with the avoidance, evasion, and real distortions generated by the regime’s preferential treatment.

This extensive-margin rationale connects our paper to a large development literature studying policies intended to bring small firms into the formal economy. Early evidence showed that reducing one-time registration costs, providing information, or simplifying registration often produces only modest increases in formalization, suggesting that for many informal firms the private benefits of formality are insufficient to compensate for its recurring costs ([De Mel, McKenzie, and Woodruff, 2013](#); [Bruhn and McKenzie, 2014](#); [De Andrade, Bruhn, and McKenzie, 2016](#); [Ulyssea, 2020](#)). Other interventions that increase the benefits of registration or substantially reduce ongoing tax and compliance costs can generate larger effects ([Benhassine et al., 2018](#); [Farias and Hsu Rocha, 2026](#)). Importantly, however, additional registration does not mechanically imply firm growth, higher productivity, or greater tax revenue. We contribute to this literature by moving beyond the number of firms induced to register. We characterize the marginal firms attracted by the simplified regime and study their counterfactual choices. Some induced firms come from the outside option and therefore represent additional business registration, whereas others would have operated under the alternative tax regime and primarily represent a reallocation of activity across tax regimes. We further show that marginal participation can be driven not by the production characteristics targeted by the policy but by firms’ ability to exploit tax-planning opportunities embedded in the regime.

Our findings also relate closely to the literature on income shifting by business owners. When labor and capital income or corporate and personal income face different tax treatments, owner-managers can respond to taxes by changing the form in which a given amount of economic income is reported rather than by changing real economic activity ([Gordon and Slemrod, 1998](#)). This margin has been particularly well documented in Nordic dual-income-tax systems. [Fjaerli, Lund, et al., 1997](#) study owners’ choice between wages and dividends in Norway; [Pirttilä and Selin, 2011](#) document shifts between labor and capital income following the introduction of Finland’s dual income tax; [Le Maire and Schjerning, 2013](#) show that income shifting explains a substantial portion of taxable-income bunching among Danish self-employed individuals; [Alstadsæter and Jacob, 2016](#) document large shifting responses to dividend taxation among Swedish owner-managers; and [Harju and Matikka, 2014](#) separately identify real and income-shifting responses among Finnish business owners. Relatedly, [Tazhitdinova, 2020](#) shows that tax differences across organizational forms affect both income shifting and business entry. Our setting introduces a different interaction between business and personal tax bases. Under Factor R, reporting owner income as wages rather than dividends not only reallo-

cates income across personal tax bases: because owner wages enter the payroll-to-revenue eligibility criterion, the shifting decision can move the entire firm to a substantially lower business-tax schedule. We exploit this feature to compare income shifting directly with alternative ways of satisfying the rule, including increasing employee payroll and reducing reported revenues. We find that incumbent firms overwhelmingly adjust through owner compensation rather than the intended employment margin. Moreover, the same ability to shift owner income affects selection into the simplified regime: marginal entrants are distinguished from inframarginal firms by owner wages rather than by employee payroll. Income shifting therefore matters not only for behavior within the regime, as in much of the existing literature, but also for the composition of firms attracted by preferential business taxation.

Finally, our paper contributes directly to a growing literature on the design of simplified and dual-regime tax systems. [Bruhn and Loeprick, 2016](#) provide early evidence that preferential small-business regimes can simultaneously increase formal firm creation and reduce tax compliance, illustrating that gross formalization gains need not translate one-for-one into gains in reported activity. [Wei and Wen, 2024](#) formalize a related trade-off: a preferential turnover-tax regime can induce informal firms to enter the tax system while encouraging firms near its upper eligibility threshold to strategically reduce reported sales. [Hoy et al., 2024](#) document further trade-offs in simplified-regime design using administrative and survey evidence from Kenya, while [Sharma et al., 2025](#) study the optimal design of systems that combine turnover and profit taxation. [Agostini et al., 2018](#) provides descriptive evidence that high-income individuals in Chile are the main users of special tax regimes for small firms, possibly affecting horizontal equity. Closest to our paper, [Gravoueille and Stantcheva, 2025](#) study the choice of simplified tax regimes by self-employed individuals in France and quantify the extent to which taxpayers' attraction to these regimes reflects a preference for simplicity versus opportunities for tax evasion. We address a complementary question: rather than decomposing the value taxpayers attach to simplicity, we study how the design of a preferential simplified regime affects both the behavior of participating firms and the composition of firms selected into it. Factor R makes eligibility depend on an endogenous production-and-reporting ratio, allowing us to distinguish firms that benefit because they are naturally labor intensive from firms that can manufacture eligibility through tax planning. Combining this variation with linked firm, owner, payroll, and third-party data, we identify the margins through which incumbent firms obtain preferential treatment, characterize the firms induced to enter the regime, and recover whether those marginal entrants would otherwise have remained outside the formal tax regime or operated under an alternative tax system. The results show that evaluating simplified tax regimes requires accounting jointly for the firms they

attract and for the behavioral margins through which those firms respond to the tax preference.

2 Context and Data

Brazil is one of the largest middle-income countries with more than 220 million inhabitants. The production landscape is dominated by small and medium-sized firms, which accounted for more than 80% of the approximately 7 million active firms in 2022. SMEs are also important for employment as they comprise 31% of formal employees and 26% of total payroll. However, they represent a much smaller share of the government’s revenue from business taxation (8%).

Table 1: Totals by Tax Regime (2022)

	Firms	Tax Payments (in millions R\$)	Employees	Payroll (in millions R\$)
Lucro Real	220,384	1,354,174	34,126,577	1,584,977
Lucro Presumido	1,073,033	201,994	7,745,465	290,290
Simples Nacional	5,560,139	150,618	19,000,890	660,637
MEI	9,429,910	253,050	5,185	2,483
Totals	16,283,466	1,711,972	61,134,749	2,538,389
Totals – MEI	6,853,556	1,706,787	60,872,933	2,535,905
Share Simples (excl. MEI)	81.13%	8.08%	31.21%	26.05%

Notes: This table reports general statistics regarding the Brazilian tax system. The rows refer to the alternative tax regimes, and the columns show their relative importance in terms of number of firms, tax payments, employment, and payroll.

Table 1 provides descriptive statistics for firms in each tax regime. The largest firms are in *Lucro Real*, a standard profit tax regime mandated for firms with annual revenues above R\$78 million. The marginal tax rate on profits ranges from 15 to 35%. The second tax regime is *Lucro Presumido*, which also covers large firms as those with annual revenues above R\$4.8 million must opt in either for *Lucro Presumido* or *Lucro Real*. The critical difference is that firms in this regime pay taxes on the presumptive profits. Common to both tax regimes is the requirement that firms submit several forms to the tax authority, including not only complete digital accounting records and fiscal bookkeeping but also state and municipal tax forms. SMEs fall under *Simples Nacional*, with a R\$4.8 million threshold. This is a simplified tax regime, under which taxpayers submit a single, straightforward tax form to the tax authority to report basic information. SMEs are taxed on revenue, with the rate depending on revenue level and sector. Finally, low-income self-employed can opt-in for *Microempreendedor Individual* (MEI), as long as their annual revenues do not

exceed R\$81,000. This is also a simplified tax regime with a flat payment, and it is not legally allowed to have more than one employee. It was designed to increase registration of informal employees (Farias and Hsu Rocha, 2026).

This paper focuses on *Simples Nacional*, the cornerstone of SME policy. It was established in federal law in 2016, and it covers a heterogeneous range of firms, from high-income individuals providing consulting services to medium-sized labor-intensive manufacturing firms. Moreover, owners of firms in *Simples Nacional* rank between the 95th and 99.5th percentiles of the income distribution in the country (Palomo et al., 2025).

Within *Simples Nacional*, firms are currently assigned to one of the five sector-specific “Anexos” (schedules) with progressive marginal rates by gross-revenue brackets. The underlying principle of the Anexos is that different sectors have different profit margins, thereby making explicit the presumptive component of this simplified tax regime. Because the service sector is highly heterogeneous, Anexos III, IV, and V differentiates among activities within the service sector. Moreover, to further approximate true profit margins, some of these service-related activities are subject to the *Factor R* rule. This rule states that the corresponding Anexo (and therefore the tax schedule) depends on the firm’s payroll-to-revenue ratio. Firms with large Factor R are assigned to Anexos with lower revenue tax rates, and vice versa. The Online Appendix presents all tax schedules corresponding to *Simples Nacional*.

The tax regime of *Lucro Presumido* also deserves further explanation, as it is the main alternative to *Simples Nacional*, and, therefore, the key counterfactual we investigate. In there, firms pay corporate taxes on a statutory measure of presumed profits rather than on their actual profits. The presumed profit margin is a fixed share of gross revenue that depends on the firm’s activity. For most commercial and industrial activities, the relevant margins are relatively low—around 8–12 percent of revenue—whereas for most service activities the presumed profit margin is 32 percent. Corporate taxes are then applied to these presumed profits, with a standard combined tax rate of 24 percent.³ Payroll taxes and state or municipal taxes are assessed separately, as they are not part of a simplification regime. During our study period, firms could opt for *Lucro Presumido* if their previous-year revenues did not exceed R\$78 million, subject to restrictions for certain activities and types of firms.

Lucro Presumido therefore differs substantially from *Simples Nacional*. First, its revenue ceiling is much higher: R\$78 million compared with R\$4.8 million under *Simples Nacional*. Second, while *Lucro Presumido* determines corporate income taxes by applying activity-specific presumed profit margins to revenue and requires firms to calculate

³Corporate profit taxation consists of the *Imposto de Renda da Pessoa Jurídica* (IRPJ), levied at a standard rate of 15 percent, and the *Contribuição Social sobre o Lucro Líquido* (CSLL), generally levied at 9 percent.

the different taxes separately, Simples Nacional taxes firms more directly on gross revenue through a unified, progressive schedule whose rates depend on the firm's activity and revenue. Simples also consolidates most federal taxes, employer social-security contributions, and state or municipal taxes into a single payment, whereas these obligations are generally paid separately under Lucro Presumido. Thus, Simples provides both preferential taxation and substantial administrative simplification for small firms, while Lucro Presumido is primarily an alternative method of calculating taxable corporate profits and remains available to considerably larger firms.

The Factor R Reform

On 1 January 2018, the Brazilian government introduced a reform on how the Factor R policy works. Before, a very restricted set of activities were subject to the Factor R rule, and the system was extremely complex. For instance, there were eight payroll-to-revenue ratio thresholds that implied very small discontinuities in the average and marginal revenue tax rates, and these discontinuities varied across the more than 20 revenue brackets. Following 2018, the government adopted a clear Factor R rule. Firms with a payroll-to-revenue ratio above 28% would be assigned to the tax schedule in Anexo III, while those below would go to Anexo V.

Three elements are critical to understand how the reform changed firms' incentives. First, there is a large group of service-related activities that were not subject to any Factor R rule and the reform included them. These were firms listed in the previously existing Anexo VI, including activities such as engineering services, consulting, and journalism, among others (see Table A.1 for details). Second, the distinction between Anexo III and Anexo V entails a substantial drop in the average tax rate, creating a tax notch and strong incentives to cross the 28% threshold.⁴ Finally, Table A.4 firms previously in Anexo VI were facing revenue tax rates closer to the ones in the new Anexo V. This means that firms that had a pre-policy payroll-to-revenue ratio above 28% faced a plain revenue tax cut.

The last component of the reform worth mentioning is what it did not modify, since that provides a control group for a clean identification strategy. Activities in Anexo IV were not subject to Factor R, and they remained unaffected. These activities include construction, logistics, landscaping, interior design, security, cleaning, and legal services.

Finally, as the Factor R could be increased through owner wages, it is worth mentioning the interaction with other tax payments. In Brazil, when the owner pays himself in wages (*prolabore*), she must pay two wage taxes. The first is the contribution to the social security system (INSS) of about 11%. This payment is linked to social security benefits, such as retirement pension and life insurance for the family. The relative value of ben-

⁴Reporting additional payroll entails social security contributions and, potentially, labor income taxes if the owner pays himself wages. Taking this into account, we show that the effective tax rate still drops substantially above the threshold.

efits over contributions explains why a significant share of firm owners pay themselves wages, even in the pre-policy period, when the dividend tax is zero. Figure A.1 shows that the tax notch is still substantial even after accounting for contributions.⁵ Second, owner wages are taxable income (different from dividends), and subject to the individual income tax.⁶ However, the exemption threshold for the individual income tax is very large. Therefore, the level of owner wages that must be reported in order to offset the gains from the revenue tax cut is not binding for 95% of firms in Simples Nacional.

2.1 Data

This paper uses detailed Brazilian tax data that were previously unavailable for research purposes. This is possible thanks to a collaboration between researchers at the International Tax Observatory (ITO) and *Receita Federal de Brasil* (RFB). Given the stringent confidentiality policies, all data processing has been conducted on RFB's premises by a tax auditor. We work with the following databases:

PGDAS:

It is the monthly tax declaration and payment platform for firms in the simplified business tax regime, filed with the federal tax authority and used to compute the firm's tax due under that regime. It covers essentially all opted-in CNPJs and their self-reported monthly tax declarations used to compute liabilities under the regime. Each record is tied to a firm identifier (CNPJ) and a reference month, and reports the tax base and payments that determine the Simples "DAS" due—most importantly gross revenues by month and the information needed to place the firm in the relevant schedule (Anexo) and bracket (with the rolling 12-month revenue concept used in the rate formula). The data include regime classifications and related inputs (e.g., activity/schedule and, when relevant, payroll and the payroll-to-revenue criterion that governs eligibility for Factor R schedules). Reporting is monthly and near-real-time, making it well-suited for studying behavioral responses at policy thresholds and around implementation dates.

GFIP:

It is best thought of as a monthly employer payroll report—not a standard business tax return. Firms submit it to the social security / payroll contribution administration to report employee links and earnings, which are used to assess and collect payroll-based contributions (and, in practice, it also serves as the reporting channel for a mandatory severance

⁵Note that it becomes a lower bound of the true changes in incentives as it does not count the valuation of the benefits that comes with the contribution payments.

⁶IRPF (*Imposto de Renda sobre Pessoa Física*)

savings fund). Each record is linked to an employer identifier (CNPJ/CEI) and a reference month, and contains employment and earnings information used to calculate and remit payroll-based obligations—typically including the number of workers, total wage bill (by contribution base), and the contribution amounts due/paid to social security and associated programs. Reporting is monthly and tied to compliance: firms must submit GFIP to regularize their payroll contributions, and the data therefore allow us to observe both real-side labor outcomes (formal employment and payroll levels) and filing behavior (whether/when firms report).

3 Descriptive Evidence: Bunching

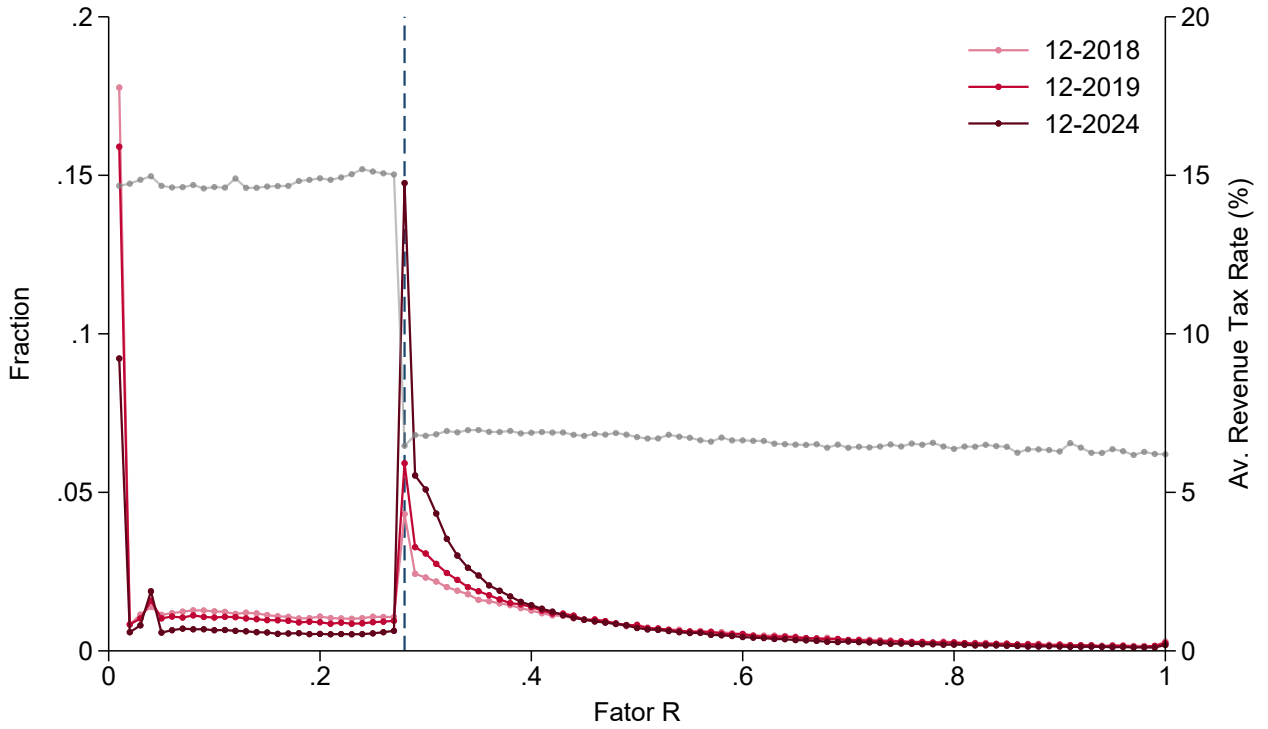
We begin by presenting a descriptive analysis showing that the Factor R policy strongly affected the distribution of the payroll-to-revenue ratio. For firms subject to the Factor R rule, Figure 1 shows the empirical average revenue tax rate. As expected, when firms cross the 28% threshold, they face a significantly lower rate, as they are moved to Anexo III.⁷

Consistent with a large tax notch, Figure 1 documents a large bunching response with about 15% of all firms subject to Factor R reporting a payroll-to-revenue ratio exactly at 28%. While tax notch theory predicts this behavior (Kleven, 2016), most empirical analyses fail to find such a clean pattern and attribute the difference to adjustment frictions (Bachas and Soto, 2021; Lobel, Scot, and Zúniga, 2024). We interpret our findings in a high-stakes context where adjustment costs are low.

Two additional patterns are worth highlighting. First, there is a hole in the distribution right below the threshold, consistent with a dominated region induced by the tax notch, suggesting that most of the bunching comes from below 28%. Second, there is an interesting dynamics in the bunching pattern that persistently grows year after year. Differentiating whether this is caused by incumbent firms learning over time or by new entrants is critical to understand firms' counterfactual and, therefore, the policy effect.

⁷The implementation of the policy was straightforward, as the system would report firms' tax due during the tax form completion. Indeed, firms could play with reported numbers and see different tax bills, making the manipulation of the system relatively easy.

Figure 1: Factor R distribution over time



Notes: This figure shows both the probability density function of the Factor R variable (left axis) and the discontinuity in revenue tax rates (y-axis). Red lines show the fraction of all firms subject to the Factor R rule at each possible Factor R value. Each red line represents a different time period. The gray line plots the mean of the average revenue tax rate paid by firms in each Factor R bin. The blue dashed line indicates the 28% threshold. Bins are of .01 size.

4 Conceptual Framework

We develop a common framework centered on the value that Factor R adds to operating under Simples. The framework serves three purposes. First, it separates the benefit of the lower revenue-tax schedule from the cost of satisfying the eligibility requirement, allowing us to characterize when firms find it profitable to obtain the tax preference. Second, it shows how heterogeneity in technology and scale determines which firms benefit most: some firms naturally satisfy Factor R because of their labor intensity, while others may find it cheap to manufacture eligibility by changing reporting or compensation choices. Third, the same value change disciplines both sides of our empirical analysis. For incumbents, it determines whether firms cross or bunch at the threshold and which margins they use to do so. For potential entrants, it determines which firms are induced to select into Simples and helps characterize the tax-regime counterfactual of those marginal firms.

4.1 Owners' Problem

Consider an owner who chooses an allocation $x_i = (e_i, L_i, \tilde{R}_i, w_i^o)$, including a composite input e_i , hired labor L_i , reported revenue $\tilde{R}_i$, and owner income reported as wages (*pro-labore*), w_i^o . The input e_i should be interpreted as a composite of productive inputs other than hired labor that are not separately modeled. Let the production technology be general and firm-specific, generating true revenues of the form:

$$R_i = A_i F_i(L_i, e_i). \quad (1)$$

A_i is a standard productivity shifter governing the firm's scale. Note that F_i is indexed by i , as firms may differ in several technological aspects, a common feature among SMEs. Firms with the same level of revenues can have substantially different labor shares, capital shares, profit margins, etc.

Employee payroll is wL_i , where w denotes the unit cost of Factor-R-eligible employee payroll. Total payroll entering Factor R is

$$P_i = wL_i + w_i^o, \quad (2)$$

and the firm's Factor R is

$$r_i \equiv \frac{P_i}{\tilde{R}_i} = \frac{wL_i + w_i^o}{\tilde{R}_i}. \quad (3)$$

Let H denote the high-tax schedule, corresponding to Anexo V, and L the favorable low-tax schedule, corresponding to Anexo III. Let $\mathcal{T}_H(\tilde{R})$ and $\mathcal{T}_L(\tilde{R})$ denote total revenue-tax liabilities under the two schedules. Over the relevant range of revenues,

$$\mathcal{T}_L(\tilde{R}) < \mathcal{T}_H(\tilde{R}) \quad \text{for } \tilde{R} > 0. \quad (4)$$

Conditional on schedule $j \in \{H, L\}$, the owner's payoff is

$$u_j(x_i) = R_i - \mathcal{T}_j(\tilde{R}_i) - wL_i - v(e_i) - \Phi(L_i) - c(R_i - \tilde{R}_i) - \Gamma(w_i^o) \quad (5)$$

The function $v(\cdot)$ captures the cost of composite input e_i .⁸ $\Phi(\cdot)$ captures other costs associated with hired labor and may include a fixed hiring cost.⁹ The function $c(\cdot)$ is the cost of underreporting revenue, and $\Gamma(\cdot)$ is the cost of reporting owner income as wages rather than dividends. The latter includes payroll contributions and personal income taxes, as well as tax-planning costs and the risk of penalties.

⁸For example, when e_i represents owner effort, this includes the opportunity or disutility cost of the owner's time.

⁹This is not strictly necessary for the main implication of the conceptual framework, but it is helpful to map the share of firms with no employees.

Before the reform, affected firms faced the high schedule H . After the reform, they continue to face H if $r_i < T$, but receive the favorable schedule L if

$$r_i \geq T, \quad T = 0.28. \quad (6)$$

Thus, the reform does not simply lower the revenue tax. It makes access to the lower schedule conditional on an endogenous statistic that firms can alter through hired labor, owner compensation, reported revenue, or changes in production.

4.2 The Factor R Value

Define the firm's maximized payoff under the high schedule as

$$\Pi_H \equiv \max_{x_i} u_H(e_i, L_i, \tilde{R}_i, w_i^o), \quad (7)$$

and its unconstrained maximized payoff under the low schedule as

$$\Pi_L \equiv \max_{x_i} u_L(e_i, L_i, \tilde{R}_i, w_i^o). \quad (8)$$

The latter is a hypothetical object in which the firm receives the favorable schedule regardless of its Factor R. It isolates how the firm would re-optimize in response to the lower revenue tax in the absence of an eligibility requirement.

Let $x_i^L = (e_i, L_i, \tilde{R}_i, w_i^o)$ denote an allocation attaining (8)¹⁰, and define

$$r_i^N \equiv r(x_i^L). \quad (9)$$

We refer to r_i^N as the firm's *natural Factor R under the favorable schedule*: it is the payroll-to-revenue ratio the firm would choose after fully re-optimizing to the low-tax schedule if eligibility did not depend on Factor R.

Finally, define the value of the best allocation that actually satisfies the eligibility condition:

$$\Pi_T \equiv \max_{x_i: r(x_i) \geq T} u_L(e_i, L_i, \tilde{R}_i, w_i^o). \quad (10)$$

It is useful to decompose the value generated by Factor R into two objects. First, define the unconstrained benefit of receiving the favorable tax schedule:

$$B \equiv \Pi_L - \Pi_H, \quad B > 0 \quad \forall i \text{ with } R_i > 0 \quad (11)$$

¹⁰If the unconstrained low-tax problem admits multiple optimal allocations, we assume that they generate the same employee-payroll share and owner-wage share, and hence the same Factor R. This allows the allocation itself to be non-unique while making p^N , ω^N , and r^N uniquely defined

Second, define the optimized cost of satisfying the Factor R eligibility restriction:

$$C \equiv \Pi_L - \Pi_T \geq 0. \quad (12)$$

The object C is an opportunity cost, measured in optimized firm payoff. It compares the allocation the firm would ideally choose under the favorable tax schedule with the best allocation available under that same schedule once the Factor R restriction is imposed. Because the two problems differ only by the eligibility restriction, C isolates the loss generated by that restriction. Importantly, it is not the cost of any particular manipulation: the firm can re-optimize jointly over hired labor, the composite input e_i , owner compensation, and revenue reporting.

The actual increase in the firm's operating value generated by Factor R is therefore

$$\begin{aligned} G &\equiv \max \{ \Pi_H, \Pi_T \} - \Pi_H \\ &= [B - C]_+. \end{aligned} \quad (13)$$

Equation (13) is the central object of the framework. The reform increases the value of Simples for a firm when the benefit of receiving the favorable tax schedule exceeds the minimum loss required to satisfy the eligibility rule. If the firm's unconstrained low-tax allocation already satisfies Factor R, then imposing the restriction does not alter its preferred allocation. On the contrary, when the eligibility restriction excludes the firm's preferred low-tax allocation, the firm benefits from Factor R only if the tax benefit is sufficiently large to compensate for this distortion.

4.3 Which firms gain from Factor R?

Equation 13 shows that heterogeneity in the value of Factor R can arise through two conceptually distinct mechanisms. Some firms naturally generate a sufficiently high payroll-to-revenue ratio and therefore receive the full benefit of the lower tax schedule. Other firms would naturally locate below the threshold but can profitably modify their behavior to satisfy it. We refer to these mechanisms as *natural eligibility* and *manufactured eligibility*, respectively.

4.3.1 Natural eligibility and labor intensity

We first consider heterogeneity in firms' natural labor intensity. Recall that x_i^L denotes the allocation firm i would choose under the favorable tax schedule if access to that schedule did not depend on Factor R. Define its natural employee-payroll intensity as

$$p_i^N \equiv \frac{wL_i^L}{\widetilde{R}_i^L}, \quad (14)$$

and its natural owner-wage share as

$$\omega_i^N \equiv \frac{w_i^{o,L}}{\widetilde{R}_i^L}. \quad (15)$$

These are equilibrium objects resulting from the firm's production technology and its optimal choices under the low-tax schedule. In particular, p_i^N summarizes the heterogeneity in labor intensity. Then, the firm's natural Factor R can then be written as

$$r_i^N = p_i^N + \omega_i^N. \quad (16)$$

A firm is *naturally eligible* if its unconstrained low-tax allocation already satisfies the statutory requirement,

$$r_i^N \geq T. \quad (17)$$

Therefore, a higher natural employee-payroll share makes natural eligibility more likely and may be sufficient to qualify for the low tax schedule whenever $p_i^N \geq T$.

Natural eligibility has an important implication for the reform's value. Because x_i^L already satisfies the eligibility requirement, imposing Factor R removes no allocation that the firm would otherwise choose. Thus, for naturally eligible firms, Factor R operates exactly as an unconstrained reduction in the revenue tax.

4.3.2 Manufactured eligibility

Consider instead a firm whose unconstrained low-tax allocation does not satisfy Factor R:

$$r_i^N < T. \quad (18)$$

Such a firm can obtain the favorable schedule only by departing from its preferred low-tax allocation. It may increase employee payroll, shift owner income from dividends to wages, reduce reported revenue, alter its production choices, or combine these margins. The optimized value loss from doing so is precisely Equation 12.

Manufacturing eligibility is profitable if and only if

$$B > C. \quad (19)$$

To characterize how this comparison varies with potential scale A_i , define, for firms

satisfying (18),

$$Q \equiv \frac{C}{B}. \quad (20)$$

Assumption 1 (Scale single crossing in manufactured eligibility). *Fix the firm's non-scale technology F_i , and consider a connected interval of productivity levels A over which the firm does not naturally qualify:*

$$r_i^N(A) < T.$$

On this interval,

$$Q_i(A) \equiv \frac{C_i(A)}{B_i(A)}$$

is continuous and strictly increasing in A , and takes values both below and above one.

Assumption 1 requires the optimized cost of satisfying Factor R to increase with potential scale faster than the benefit of the favorable tax schedule. Importantly, all endogenous choices are allowed to adjust with A , including the firm's natural Factor R. Thus, the assumption does not compare firms at a fixed distance from the eligibility threshold. Instead, the assumption allows the firm's natural Factor R to change with scale, but it requires the scale force to dominate any endogenous change in the amount of adjustment required to qualify.

The intuition is especially transparent when the cheapest adjustment is owner-income shifting. Suppose, only for illustration, that a non-qualifying firm holds the other components of its unconstrained low-tax allocation fixed and closes the Factor R gap by reporting additional owner wages. The required additional owner wage is

$$m(A) = [T - r^N(A)] \tilde{R}^L(A). \quad (21)$$

For a given distance from the threshold, the amount of income that must be shifted grows with firm scale. If the private cost of shifting owner income is convex, the cost of satisfying the same payroll-to-revenue requirement can therefore rise more than proportionally with scale. Assumption 1 requires this force to dominate any corresponding increase in the value of the tax cut.

Proposition 1. *Natural and manufactured eligibility*

1. **Natural eligibility.** *If the firm's unconstrained low-tax allocation satisfies*

$$r_i^N \geq T, \quad (22)$$

then the eligibility restriction does not constrain the firm's preferred allocation. Therefore,

$$C_i = 0, \quad G_i = B_i > 0. \quad (23)$$

In particular, $p_i^N \geq T$ is sufficient for natural eligibility.

2. **Manufactured eligibility.** If the firm's unconstrained low-tax allocation satisfies $r_i^N(A) < T$ and Assumption 1 holds, there exists a unique cutoff $\bar{A}^M(F_i)$ (for a given non-scale technology) satisfying

$$Q_i(\bar{A}^M(F_i)) = 1. \quad (24)$$

Within this non-natural-eligibility region,

$$A < \bar{A}^M(F_i) \implies C_i(A) < B_i(A) \implies G_i(A) > 0, \quad (25)$$

whereas

$$A > \bar{A}^M(F_i) \implies C_i(A) > B_i(A) \implies G_i(A) = 0. \quad (26)$$

At $A = \bar{A}^M(F_i)$, the firm is indifferent between satisfying Factor R and remaining under the high-tax schedule.

The proof is provided in the Online Appendix.

Proposition 1 identifies two distinct reasons why Factor R increases the value of Simples. First, firms whose unconstrained choices under the favorable tax schedule generate a sufficiently high Factor R are naturally eligible. For these firms, the eligibility requirement is irrelevant for behavior: they receive the full value of the lower tax schedule and choose exactly the allocation they would choose under an unconditional tax cut. Natural labor intensity is an important source of this heterogeneity, since firms with sufficiently high employee payroll relative to revenue may satisfy the requirement through employee payroll alone.

Second, firms whose natural Factor R lies below the threshold can benefit only by departing from their preferred low-tax allocation. Conditional on the firm's non-scale technology, Assumption 1 implies that manufacturing eligibility is profitable at sufficiently low levels of potential scale.

4.4 Implications for incumbent firms

Consider firms that were already operating in Simples before the reform. For these firms, participation in the regime is held fixed, and the reform changes the allocation they choose within Simples.

To characterize the firm's post-reform location in the Factor R distribution, define its optimized payoff under the favorable schedule conditional on attaining a particular Factor R:

$$W_{L,i}(r) \equiv \max_{x_i: r(x_i)=r} u_L(x_i). \quad (27)$$

By construction, $W_{L,i}(r)$ reaches its unconstrained maximum at the firm's natural Factor R , r_i^N . We impose the following regularity condition.

Assumption 2. *Factor R single-peakedness* The set of attainable Factor R values is an interval containing T , and $W_{L,i}(r)$ is continuous and strictly single-peaked at r_i^N .

The relevant implication of Assumption 2 is that when $r_i^N < T$, the firm's best allocation among those satisfying $r_i \geq T$ lies exactly at the eligibility threshold. Once the firm moves above T , moving farther above it provides no additional tax benefit and only moves the firm farther from its preferred low-tax allocation.

Let r_i^0 denote the firm's optimal Factor R before the reform, when it faces the high-tax schedule, and let r_i^1 denote its optimal Factor R after the reform.

Proposition 2 (Incumbent responses around the Factor R threshold). *Under Assumption 2:*

1. **Initially above the threshold.** If $r_i^0 \geq T$, the firm never crosses below the threshold and

$$r_i^1 = \begin{cases} r_i^N, & \text{if } r_i^N \geq T, \\ T, & \text{if } r_i^N < T. \end{cases}$$

2. **Initially below the threshold.** If $r_i^0 < T$, then

$$r_i^1 = \begin{cases} r_i^N, & \text{if } r_i^N \geq T, \\ T, & \text{if } r_i^N < T \text{ and } B_i > C_i, \\ r_i^0, & \text{if } r_i^N < T \text{ and } B_i < C_i. \end{cases}$$

When $r_i^N < T$ and $B_i = C_i$, the firm is indifferent between remaining below the threshold and bunching at T .

The proof is provided in the Online Appendix.

The first part of Proposition 2 delivers a particularly clean prediction. A firm that was already above the threshold before the reform never optimally crosses below it. The reason does not depend on the firm's technology or on how its inputs respond to the tax cut. The firm can always retain its pre-reform allocation, remain eligible, and pay strictly lower taxes at that same allocation. Hence, some qualifying allocation is strictly preferable to any allocation attainable under the high-tax schedule.

For firms initially below the threshold, obtaining the favorable schedule instead requires an upward movement in Factor R. If the firm's unconstrained low-tax optimum already satisfies $r_i^N \geq T$, it crosses the threshold without being constrained once it does so. If $r_i^N < T$, however, qualifying requires the firm to depart from its preferred low-tax allocation. Proposition 1 determines whether this manufactured eligibility is worthwhile. A firm for which $B_i > C_i$ qualifies, and because locating farther above the threshold provides no additional discrete tax benefit, it bunches at T . Otherwise, it remains below the threshold.

An important implication is that the excess mass at the threshold can originate from both sides of the pre-reform Factor R distribution. Firms initially below T can move upward to obtain the favorable schedule, while firms initially above T can re-optimize downward until the eligibility condition binds. Proposition 2 rules out only one movement: an initially above-threshold firm should not cross below the eligibility threshold.

The framework also provides a direct mapping to observable adjustment margins. Let

$$P_i = wL_i + w_i^o.$$

Any firm locating at the threshold after the reform satisfies

$$P_i^1 = T\tilde{R}_i^1.$$

Since prior to the reform

$$P_i^0 = r_i^0 \tilde{R}_i^0,$$

changes between the pre- and post-reform allocations satisfy the accounting identity

$$\Delta P_i - T\Delta\tilde{R}_i = (T - r_i^0)\tilde{R}_i^0. \quad (28)$$

For a buncher coming from below, $r_i^0 < T$, so

$$w\Delta L_i + \Delta w_i^o - T\Delta\tilde{R}_i = (T - r_i^0)\tilde{R}_i^0 > 0. \quad (29)$$

The firm must therefore close its initial distance from the threshold through some combination of higher employee payroll, higher owner wages, or lower reported revenue. Which margin moves depends on the relative economic costs of the available responses. Their observed changes therefore reveal how firms find it cheapest to satisfy Factor R.

For a buncher coming from above, the right-hand side of (28) is negative. Such a firm would like to reduce its payroll-to-revenue ratio in response to the favorable schedule, but the eligibility restriction prevents it from moving below T .

4.5 Implications for selection into Simples

We next consider potential firms choosing among Simples Nacional (S), Lucro Presumido (P), and an outside option (0). The outside option includes both inactivity and informality, as we can't empirically identify among them.

Let K_S denote the fixed cost of operating under Simples Nacional, including registration and compliance costs. Pre-reform, the value of choosing S for a given firm is:

$$V_S^0 = \Pi_H - K_S, \quad (30)$$

while after the reform,

$$\begin{aligned} V_S^1 &= \max\{\Pi_H, \Pi_T\} - K_S \\ &= V_S^0 + G. \end{aligned} \quad (31)$$

Equation (31) provides the direct connection between the intensive and extensive margins. Factor R raises the value of selecting into S by exactly the same object G that governs the incentive to obtain the favorable schedule for firms already operating in the regime.

Proposition 1 therefore has an immediate selection implication. Factor R makes S more valuable for both firms that naturally satisfy the eligibility requirement and firms that can manufacture eligibility at sufficiently low cost. Whether this increase in value actually induces a firm to select into S depends on its alternative regime choice.

Let V_P denote the value of operating under Lucro Presumido, which is unaffected by Factor R, and normalize the outside option to zero. To characterize the counterfactual regime of firms induced into Simples, we impose the following productivity-sorting condition.

Assumption 3 (Productivity sorting across regimes). *Fix the firm's non-scale technology F_i .*

1. $V_S^0(A)$ is continuous and strictly increasing in A and crosses the outside option exactly once. Let $A_E^0(F_i)$ satisfy

$$V_S^0(A_E^0(F_i)) = 0.$$

2. The relative value of Lucro Presumido,

$$D^0 \equiv V_P - V_S^0,$$

is continuous and strictly increasing in A and crosses zero exactly once at $A_{SP}^0(F_i)$, with

$$A_E^0(F_i) < A_{SP}^0(F_i).$$

The first condition captures the standard scale effect in formal entry: sufficiently low-productivity firms do not generate enough surplus to cover the fixed costs of formal operation. The second is a single-crossing condition between the two formal tax regimes. It states that Lucro Presumido becomes relatively more attractive as productivity and potential scale increase. A natural microfoundation is that Lucro Presumido entails higher fixed accounting and compliance costs than Simples but offers a relatively more attractive variable payoff at larger scale.

Assumption 3 implies the following pre-reform allocation, apart from exact indifference points:

$$j_i^0 = \begin{cases} 0, & A_i < A_E^0(F_i), \\ S, & A_E^0(F_i) < A_i < A_{SP}^0(F_i), \\ P, & A_i > A_{SP}^0(F_i). \end{cases} \quad (32)$$

Thus, conditional on the firm's non-scale technology, potential scale orders firms from the outside option, to Simples, to Lucro Presumido.

Proposition 3 (Counterfactual regime of firms induced into Simples). *Under Assumption 3, fix the firm's non-scale technology F_i and consider a firm induced by Factor R to choose Simples.*

1. *If its pre-reform counterfactual is the outside option, then*

$$A_i < A_E^0(F_i).$$

2. *If its pre-reform counterfactual is Lucro Presumido, then*

$$A_i > A_{SP}^0(F_i).$$

Since $A_E^0(F_i) < A_{SP}^0(F_i)$, conditional on the same non-scale technology, firms induced from Lucro Presumido have higher potential scale than firms induced from the outside option.

The proof is provided in the Online Appendix.

The same object G determines whether Factor R expands Simples at either margin. If Factor R raises the value of Simples for firms at the pre-reform outside–Simples boundary, the lower boundary shifts downward, inducing firms that would otherwise choose the outside option to select into Simples. Similarly, if it raises the value of Simples at the pre-reform Simples–Lucro Presumido boundary, the upper boundary shifts upward, inducing firms that would otherwise operate under Lucro Presumido. These boundary movements require the corresponding single-crossing properties to continue to hold after the reform; we state the precise conditions in the Online Appendix.

Propositions 1 and 3 therefore characterize two distinct dimensions of selection. Proposition 1 determines *why* Factor R raises the value of Simples: because the firm naturally satisfies the eligibility rule or because it can manufacture eligibility at sufficiently low cost. Proposition 3 instead determines *where* an induced firm would have operated in the absence of the reform.

Importantly, the framework does not impose a one-to-one mapping between these two dimensions. Firms induced from either the outside option or Lucro Presumido may benefit through natural or manufactured eligibility. In particular, although manufactured eligibility becomes less attractive with potential scale conditional on a given non-scale technology, the scale thresholds governing eligibility and regime choice both depend on F_i . Whether marginal entrants are selected primarily because of their natural labor intensity or because of their ability to manufacture eligibility, and whether these firms would otherwise have chosen the outside option or Lucro Presumido, are therefore empirical questions.

5 Intensive Margin

This section focuses on how incumbent firms (those in *Simples Nacional*) respond to the Factor R policy and what we can learn from such behavior. The central feature of the reform is that it created different incentives depending on firms' location in the pre-policy Factor R distribution. Firms initially above the 28% threshold became automatically eligible for a lower revenue tax rate, whereas firms initially below the threshold had to change their payroll-to-revenue ratio to obtain the tax reduction.

In what follow, we present the empirical strategy, where we compare exposed vs. non-exposed firms in a series of *difference-in-differences*. Guided by the prediction in Section 4.4, we estimate the effects on the location in the Factor R distribution, as well as the effects on all Factor R components, separately for low and high pre-policy Factor R.

5.1 Empirical Strategy

In this section, we causally estimate the effect of the Factor R policy on incumbent firms' behavior. We exploit the fact that firms with pre-policy Factor R above 28% were automatically eligible for a tax cut, while firms below 28% faced large tax incentives to increase their payroll-to-revenue ratio. Therefore, we conduct our analysis separately for firms below and above 28%, which we refer to as low and high Factor R, respectively.

Our sample is comprises of firms with positive revenues in all quarters between 2016 and 2019, covering two years before and after the reform. We define our treatment group

as firms belonging to Anexo VI for the entire pre-period, and our control group as those belonging to Anexo IV prior to the reform. Table A.1 presents the different activities for treatment and control. Both treated and control correspond to firms in the service sector. The activities in the treated group were all automatically subject to the Factor R rule, while those in the control group were not.

An important distinction between the two groups is worth highlighting. In the pre-period, they are taxed at very different revenue tax rates. Industries in the control group were assumed to have lower profit margins, and therefore faced lower tax rates. This is helpful for our identification, as firms in the control group have no incentives to relabel their industry code to take advantage of the Factor R policy. Table 2 illustrates the baseline differences between the treatment and control group.

Table 2: Descriptive Statistics: Firms in 2017-Q4

	Accumulated 12-month				Current (Quarter)			
	Low FR		High FR		Low FR		High FR	
	Treated	Control	Treated	Control	Treated	Control	Treated	Control
Revenues	308,565	543,037	220,681	513,755	28,403	51,261	20,575	47,432
Owner Wages	17,526	17,673	33,878	29,774	1,558	1,558	2,857	2,475
Employees' Payroll	23,927	38,690	70,277	220,735	2,564	4,145	7,226	22,383
1[Revenues = 0]	0	0	0	0	0	0	0	0
1[Owner Wages = 0]	0.278	0.347	0.092	0.128	0.298	0.371	0.105	0.150
1[Employees' Payroll = 0]	0.581	0.476	0.269	0.100	0.603	0.512	0.289	0.122
1[Total Payroll = 0]	0.182	0.221	0.000	0.000	0.198	0.244	0.003	0.004
N Firms	19,863	29,594	16,320	17,692	19,863	29,594	16,320	17,692

Notes: This table reports descriptive statistics for firms observed in 2017-Q4. Low FR and High FR refer to firms classified according to their pre-reform Factor R. Treated firms are those affected by the reform, while control firms belong to the comparison group. Monetary values are in Brazilian reais.

The identification strategy is a simple *difference-in-differences* comparing treatment and control groups before and after the reform. We use a Poisson model as most outcomes can be zero for a significant share of firms.¹¹¹²

$$y_{iq} = \alpha_i + \lambda_q + \sum_{k \neq -1} \beta_k 1[\text{Treated}_i = 1] 1[q = k] + u_{iq}, \quad (33)$$

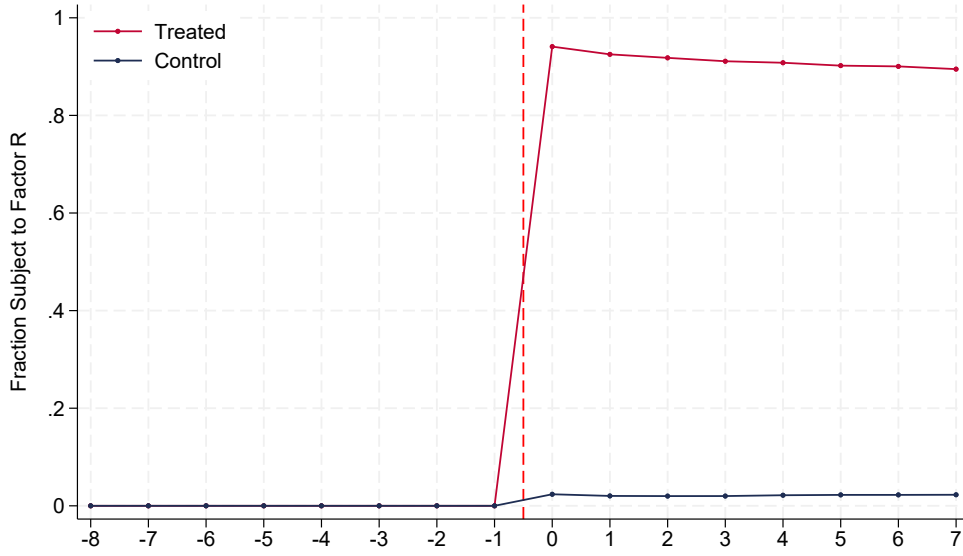
¹¹For example, Table 2 shows that 58% of low-Factor R firms in the treated group and 48% in the control have no employees.

¹²Figures A.2 and A.3 in Appendix A.1 provides the treated and control groups' raw trends for all outcomes.

where y_{it} represents the outcome, α_i is the firm FE, and λ_q is the quarter FE. Our coefficients of interest are the β_k , which represent the effect of the policy in the log of the difference between the treatment and control groups in quarter k relative to the last quarter of 2017.

Figure 2 shows how the treatment and control groups are differently exposed to the policy. This confirms that almost all firms in the treated group became subject to the Factor R rule, while only a negligible fraction of firms in the control group did, probably due to natural change in economic activity.

Figure 2: Fraction subject to Factor R by pre-policy Anexo



Notes: This Figure shows the fraction of firms subject to the Factor R policy separately by treatment status. The red line shows the group of firms that operated in Anexo VI pre-reform, while the blue line shows those that operated under Anexo IV pre-reform. The x-axis is the quarter relative to the reform, where $t = 0$ represents the first quarter of 2018.

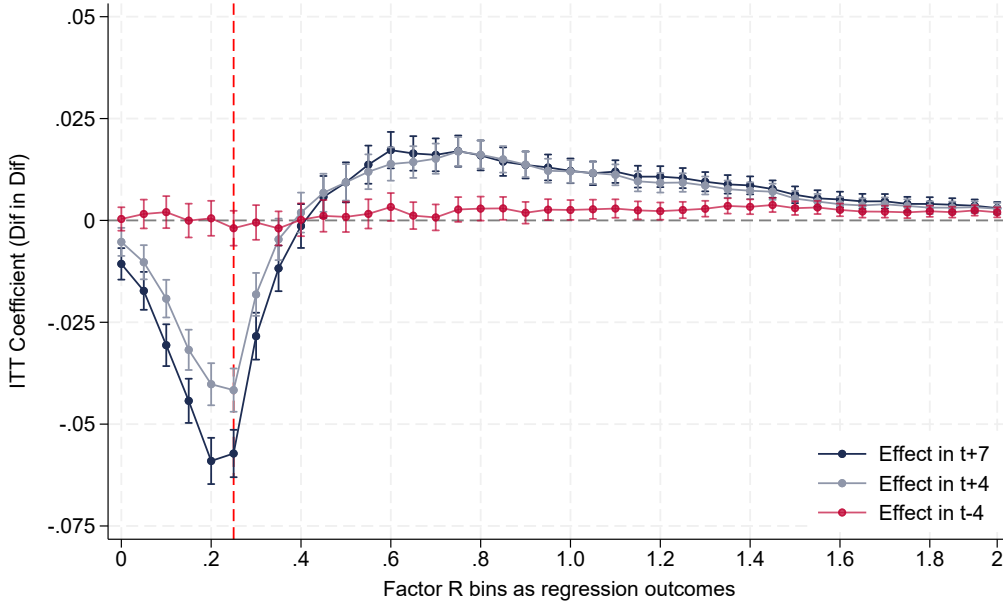
5.2 Who bunches? Location in the Factor R distribution

Proposition 2 shows that the excess mass at the 28% threshold can originate from *both* sides of the pre-policy Factor R distribution: firms initially below 28% may move up to obtain the tax reduction, while firms initially above 28% may re-optimize downward toward the threshold. This subsection provides the first empirical test of the proposition: we estimate where the firms that end up at 28% would have been located absent the reform, and we quantify how much of the bunching is drawn from each side of the threshold.

To do so, we estimate the difference-in-differences specification in equation (33) using the firm's *location* in the Factor R distribution as the outcome. We discretize the Factor R support into bins and, for each bin cutoff c , define the outcome $y_{iq} = \mathbf{1}[FR_{iq} \leq c]$, an indi-

cator for whether firm i 's Factor R in quarter q lies at or below c . This outcome is the empirical cumulative distribution function (CDF) evaluated at c , so estimating equation (33) bin by bin traces the treatment effect on the *entire* distribution of Factor R rather than on a single moment. Figure 3 summarizes these estimates. Each point on the horizontal axis corresponds to a different regression, that is a different outcome $1[FR_{iq} \leq c]$ indexed by the bin cutoff c , and the vertical axis reports the associated difference-in-differences coefficient β_k , the treatment effect on the share of firms located at or below that bin. Each line collects the β_k for a given event time k : we plot the pre-reform placebo at $t - 4$ together with the post-reform effects at $t + 4$ and $t + 7$. A negative coefficient at cutoff c means the reform reduced the mass of treated firms located at or below c ; a positive coefficient means it raised it.

Figure 3: Effect on the Factor R Cumulative Distribution Function



Notes: This figure shows the reduced form coefficients (β_k s) from Equation 33 where the outcome $y_{iq} = 1[FR_{iq} \leq c]$ and c corresponds to each bin in the Factor R distribution. Then, each dot in the x-axis refers to a different regression. Bins are of size .05, with the first bin including the zero, and the last bin comprising any value above 2 to cover the entire population of firms. The lines plot different k s: the red line is the dif-in-dif coefficient in -4 , while the blues are post-period coefficients.

Two features of the dynamics are worth noting. First, the pre-reform line ($t - 4$) is flat and statistically indistinguishable from zero across the entire distribution, confirming that treated and control firms followed parallel paths before the policy and that the estimated effects are not driven by pre-existing differences in where firms were located. Second, the response builds up over time: the displacement of mass toward the threshold is already visible at $t + 4$ but is substantially larger at $t + 7$, consistent with the descriptive evidence that bunching grows year after year as incumbent firms learn to exploit the notch. We end the event window at $t + 7$, the fourth quarter of 2019, for two reasons. It stops the analysis

before the COVID-19 shock of 2020, which would otherwise confound the comparison of treated and control firms; and it keeps the incumbent sample large, since requiring firms to be continuously active over a longer window would progressively drop firms and erode the set of incumbents we can follow.

The shape of the curve identifies where the bunching mass comes from. Because the plotted outcome is the CDF, its slope across bins is the treatment effect on the *density* (the PDF): where the curve rises the reform pushed mass into that region, and where it falls the reform pulled mass out. The bunching at 28% therefore appears as the steep positive slope at the threshold. Reading the $t + 7$ line, the effect on the CDF turns increasingly negative as we approach the threshold from below and reaches its trough just at 28%: the reform emptied the region immediately below the threshold, as firms that would have located there moved up to bunch. Crucially, just above the threshold the curve does not return to zero but turns *positive*. This is only possible if part of the bunching is composed of firms that would otherwise have been located higher up and re-optimized downward toward the threshold.

This logic delivers a clean decomposition of the bunching mass by origin. Since treated and control firms are both distributions, total mass is conserved and the coefficient curve begins and ends at zero; the mass added by the positive slope at the threshold must therefore be exactly offset by the mass removed elsewhere. Two regions of the curve have negative slope, and hence correspond to mass being removed. Because the negative slopes must sum to the positive slope, we can attribute the bunching to each side. We find that 84% of the firms that bunch at 28% would have been located below the threshold, while the remaining 16% would have been located above it.

5.3 Behavioral Responses by pre-reform Factor R group

An important implication of the Factor R policy is that it provided two clean quasi-experiments for incumbent firms. Those firms that were already above the threshold pre-policy became automatically eligible for the tax cut. This allows us to estimate how firms respond to a reduction in the revenue tax rate in terms of input demand, reported revenues, etc. On the other hand, those firms below 28% pre-policy face incentives to reach the threshold through, potentially, different behavioral responses specified in Equation 29. Therefore, we estimate Equation 33 separately for low and high-Factor R pre-policy.

We begin by showing the effect on the average revenue tax rate. Figure 4 clearly shows how the policy affected incentives differently depending on the location in the Factor R distribution pre-policy. Those who were above 28% (High FR) face an immediate drop in the revenue tax rate of about 9pp. Those who were below 28% (Low FR) do not face a large and sudden drop, but rather an initially smaller drop which grows over time. This

is due to the fact that Low FR firms do not become automatically eligible, but more and more firms enjoy the tax cut as they cross the threshold.

Revenues

We study how the policy affected reported revenues. Figure 4b shows null effects for Low FR firms, proving that reducing reported revenues is not a cheap vehicle to increase the payroll-to-revenue. However, High FR pre-policy see a significant increase in reported revenue of about 10% eight quarters after the reform. We refer the reader to Appendix B, where we translate the reduced-form estimate to a revenue elasticity to the net-of-tax rate.¹³

The increase in reported revenues for High-FR firms is consistent with re-optimization. However, two alternative stories are consistent. A firm can either increase true economic activity and, therefore, revenues, or they can reduce revenue under-reporting. Both responses are consistent with a revenue tax cut. Below, we explore in more detail what drives the increase in reported revenues.

Employees' Payroll

We now turn to the effect on employees' payroll. As Equation 29 shows, this is an alternative for Low FR firms to reach the 28% threshold. Any adjustment in the number of employees or their wages will be reflected here. Similarly, employers could decide to register informal employees or reduce payments under the table within the firm (Feinmann, Hsu Rocha, and Lauletta, 2022). We may also expect an increase in employees' payroll for High FR firms if the increase in reported revenues is due to real activity, and the non-deductibility of labor costs was restraining labor demand considerably.

Figure 4c shows basically no effects on employees' payroll for either group. Neither Low FR nor High FR firms experienced increases statistically different from zero. For the first group, this reveals that adjusting employees' payroll is not the cheapest option to take advantage of the tax rate reduction, undermining the policy goal. The result for the High FR group is also surprising. Despite the fact that revenue taxation distorts input demand and that we find a 10% increase in reported revenues, employees' payroll does not respond to the policy. This reveals that firms are more reactive in other margins.

Owner Wages

The other important behavioral response is how firm owners adjust their payments between dividends and wages when it can affect their overall tax rate. Given that Low

¹³Such elasticity assumes no constrained firms because we cannot measure the right change in the revenue tax rate for them as we cannot isolate the shadow price of the constrain. We still consider it helpful to provide an order-of-magnitude estimate for such elasticity.

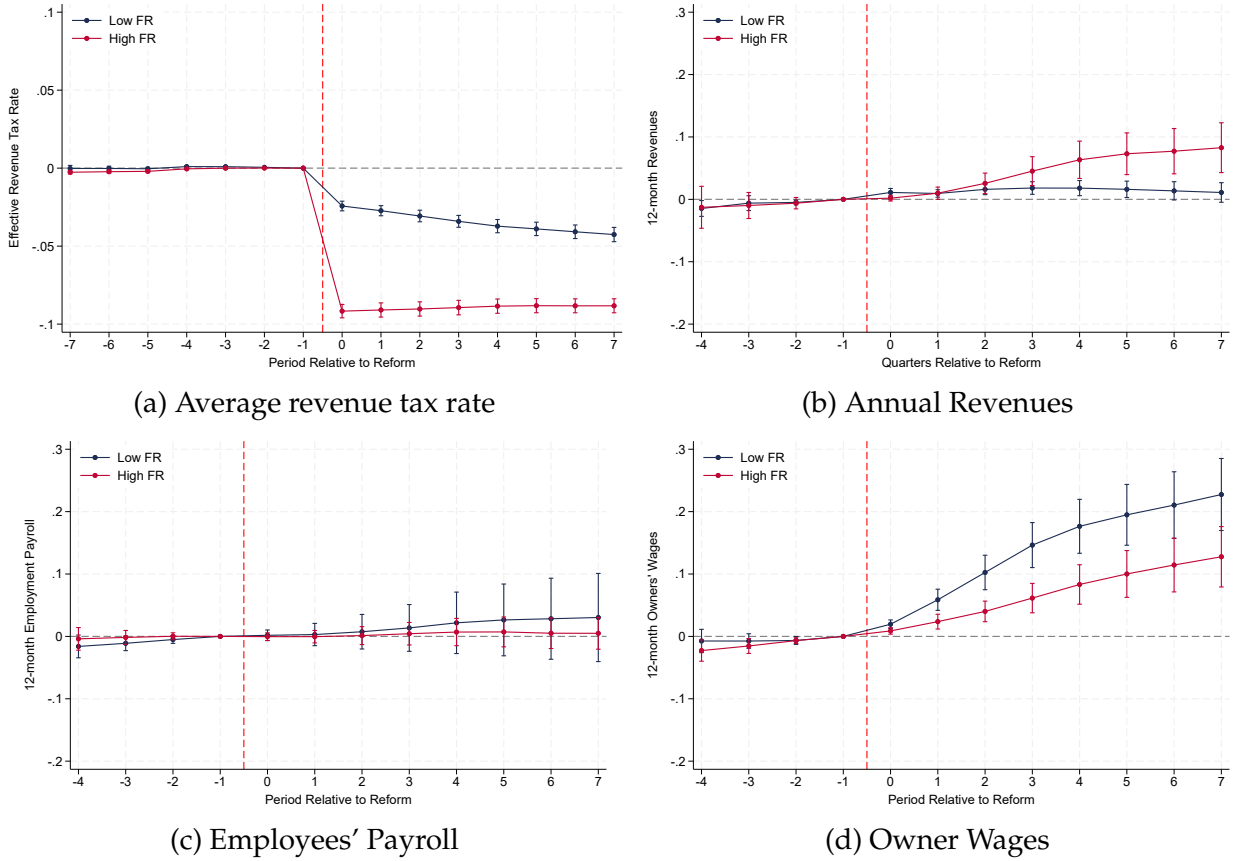
FR firms neither reduced reported revenues nor increased employees' payroll, Equation 29 implies that owners must pay themselves in wages to increase the payroll-to-revenue ratio and obtain the tax cut. For firms above 28% pre-policy, the responses depend on whether they belong to the constrained or unconstrained group, and whether the former group finds owner wages the cheapest way to relax the constraint.

Figure 4d has large and growing increases in owner wages for Low FR firms, reaching 25% two years after the reform. Interestingly, High FR firms also exhibit a significant increase in owner wages. As mentioned earlier, this may be explained by a considerable mass of constrained firms above the threshold. Figure A.4 in Appendix A.1 shows evidence that firms right above the threshold explain most of the owner wages increase, while firms far from the threshold explain most of the revenue increase.¹⁴

The comparison between the behavioral responses in owner wages and employees' payroll reveals how cheap it is for firms in simplified tax regimes to exploit loopholes. This is an important result for thinking about the proper design of such regimes, as exemptions, differential schemes, and discretionary rules may create loopholes that undermine policy goals.

¹⁴An alternative explanation (not modeled in our framework) could be that owners use some fixed rules for self-payments (e.g. 20% wages 80% dividends). If so, the increase in owner wages for High FR firms may be mechanically related to the increase in revenues. Consistently, Figure A.3d shows that the ratio of owner wages to revenues remains totally flat for High FR firms.

Figure 4: Reduced Form Estimates for Incumbent Firms



Notes: This figure shows the reduced form coefficients (β_{ks}) from Equation 33 where the outcomes are the average tax rate (Panel a), the 12-month accumulated revenues (Panel b), the 12-month accumulated employees' payroll (Panel c), and the 12-month accumulated owner wages. The outcome variable is in levels in all cases. For the average revenue tax rate, we run an OLS specification, while for Factor R components we use Poisson regressions as the distribution contains many zeros. The blue line limits the sample to firms with a Factor R below 28% pre-policy. The red line limits the sample to firms with a Factor R greater than 28% pre-policy. Vertical bars show 95% confidence intervals.

Reporting vs. Real Responses for High FR Firms

One striking result is the significant revenue response together with the null employment response for High-FR firms. This finding raises a natural question: what explains the increase in firms' revenues? To answer this question, we rely on a *production frontier analysis*. This exercise uses several variables related to firms' real economic activity to predict *true* revenues (see Section 2.1 for details). We further decompose the effect on alternative inputs.

Table 3 shows that firms' revenues estimated through predictors of true economic activity also increase, and almost in the same magnitude as reported revenues. Consistently, we observe that variable capital adjusts upward strongly after the policy. It is worth noting that the pre-period coefficient on true revenues is negative and significant, indicating a small violation of the parallel trend assumption. While the post coefficients are almost

identical to those for reported revenues, we are cautious in our interpretation and do not claim that the entire effect on revenues is real, as part of it may be explained by pretrends. However, part of it seems to be real, and the types of inputs adjusting are consistent with that.

Table 3: Reported vs. True Responses

	(1) Reported Revenues	(2) True Revenues	(3) Fixed Capital	(4) Variable Capital	(5) Current Purchases
<i>Pre-period</i>					
2016	-0.008 (0.023)	-0.033** (0.016)	0.002 (0.010)	0.021 (0.024)	-0.038 (0.025)
<i>Post-period</i>					
2018	0.045*** (0.012)	0.046*** (0.013)	0.009 (0.009)	0.053*** (0.015)	0.019 (0.019)
2019	0.083*** (0.020)	0.082*** (0.018)	0.023* (0.013)	0.065*** (0.021)	0.024 (0.024)
FE by Firm	✓	✓	✓	✓	✓
FE by Year	✓	✓	✓	✓	✓
Observations	135,864	128,228	128,228	128,228	128,228

Notes: This table presents the results of estimating Equation 33 on a series of outcomes. Column (1) shows the effect on the annual reported revenues. These estimates correspond to the coefficients for the last quarter in each year shown in Figure 4b (red line). Column (2) uses the predicted revenues based on a series of variables related to real economic activity (see Section 2.1 for details). Columns (3), (4), and (5) show the estimates for fixed and variable capital, and current purchases, respectively. The omitted year is 2017. Standard errors are reported in parentheses. All specifications are estimated using PPML and include firm and year fixed effects. The difference in observations is explained by a few firms for which we were not able to reconstruct their input variables. Standard errors are clustered at the pre-reform industry level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Extensive Margin: Firm entry

The previous section shows that Low-FR incumbents respond to the eligibility incentive predominantly through owner-income shifting rather than employee payroll or revenue reporting. High-FR firms, meanwhile, increase reported revenues following the tax cut but exhibit no detectable employment response.

While this is a very important piece of evidence on incumbent firms' behavior, it is silent about why firms select into STRs in the first place. In Section 4, Proposition 1 for-

malized that the Factor R reform made *Simples Nacional* more attractive for labor-intensive firms, and those that can cheaply manufacture their eligibility.¹⁵ Moreover, understanding the counterfactual of these groups is key, as moving firms from inactivity/informality to a STR has very different implications than switching tax regimes.

In what follows, we first explain the empirical strategy, highlighting how we address the main challenge to identify the motives for firms' selection into *Simples Nacional*. We decompose our difference-in-differences estimates into the ITT effects of new firms that would have entered regardless of the policy (*always-takers/inframarginals*) from the selection of firms that entered due to the Factor R rule (*compliers/marginals*). Second, we study firms' counterfactuals regarding alternative tax regimes, industry and contract relabeling, and real business registration.

6.1 Empirical Strategy

We leverage the fact that eligibility to the Factor R policy depended on firms' economic activity. This means that some industries were more exposed to the policy than others. The law establishes which activities are covered by each Anexo.¹⁶ Unfortunately, firms do not report their economic activity in the tax form, but only which Anexo corresponds to them. Therefore, we will use the 7-digit industry code (CNAE), reported at the registration stage, to map the economic activity defined in the law.¹⁷ For non-exposed industries, we follow the analogous approach, mapping 7-digit CNAE to Anexo IV.¹⁸

We then perform a difference-in-differences analysis at the industry level, with outcomes constructed from the cohort of new firms in each year. A new firm is defined as having positive revenues at some point during the registration year. This allows us to capture a meaningful dimension of entry, as many small firms may register but never carry out economic activity. The regression equation for the dynamic difference-in-differences looks as follows:

¹⁵Inactivity and informal firms are equivalent for the purpose of the model. Lucro Presumido is a presumptive profit tax regime with a revenue tax depending on the activity. However, it has no simplification whatsoever, meaning that payroll taxes, state taxes, and others are not unified. We provide a full characterization in Section 2.1.

¹⁶Art. 18 de la Lei Complementar 123/2006. See Appendix for full description.

¹⁷Reported CNAEs must be updated if firms change their main activities. For our purposes, the enforcement of updates is irrelevant, as we focus on new entrants.

¹⁸For robustness, we also perform a data-driven approach to define different groups of exposed and non-exposed industries. For the first group, we select industry codes that have a large share of all firms in Anexo VI during the pre-policy period. For the latter, we do the same, but based on the shares of all firms in Anexo III and IV pre-reform, thereby allowing us to verify that new firms are not merely relabeling industry codes. Figure A.6 shows that the policy did not reduce the number of new firms in any subgroup of economic activities related to Anexo III pre-policy.

$$y_{ct} = \alpha_c + \lambda_t + \sum_{k \neq 2017} \beta_k \mathbf{1}[\text{Treated}_c = 1] \mathbf{1}[t = k] + u_{ct} \quad (34)$$

Where y_{ct} is the outcome of interest for the industry code c , in year t . We include fixed effects by industry (α_c) and time (λ_t). β_k measures the gap between exposed and non-exposed industries relative to the 2017 gap. Figure A.5 in Appendix A.1 shows the raw trends, verifying the responses are driven by the treated industries rather than the control. Importantly, β_k does not directly address our main question: whether the additional firms induced by the policy entered because they are labor-intensive or because they can manufacture their eligibility through income shifting. The reason is that β_k mixes compliers' characteristics with the behavioral responses of always-takers to the policy (those firms that would have entered Simples Nacional regardless of the policy). We leverage our results for incumbent firms to separately identify these two components.

Marginal Entrants vs. Inframarginal Behavioral Responses

As we are interested in understanding the selection of new firms induced by the policy, it is important to interpret β_k precisely. After the reform, there are two groups of new firms: the ones that entered because of the policy (*marginal entrants* or *compliers*) and those that would have entered anyway (*inframarginal entrants* or *always-takers*). While the latter group is inframarginal to the entry decision, they may change their behavior under the Factor R rule. This means that β_k captures a combination of the selection of new firms and the behavioral responses of always-treated firms. Formally:

$$\beta^{DiD} = \underbrace{\left(\mathbb{E}[Y(1) | AT] - \mathbb{E}[Y(0) | AT] \right)}_{(I)} + \underbrace{\pi_C \left(\mathbb{E}[Y(1) | C] - \mathbb{E}[Y(1) | AT] \right)}_{(II)}. \quad (35)$$

Term (I) is the intensive-margin response of firms that would have entered regardless. Term (II) is the contribution of the composition effect to the reduced-form effect. Dividing it by the complier share (π_C) delivers our main object of interest: the selection gap. It reveals how those new entrants induced by the policy differ from those firm that would have entered anyway. Their differences in behavior and characteristics shed light on the entry motives. See Appendix E for details on the derivation of Equation 35.

Estimating π_C is straightforward, as it comes from using the log of total new firms as the outcome variable. However, recovering the selection gap requires knowledge of the change in behavioral responses among inframarginal entrants. Note that this is a setting with large changes in incentives, so we expect firms to respond accordingly. Also note that Term (I) cannot be estimated, as always-takers are not directly observed in the data.

Fortunately, in Section 5 we estimated the behavioral responses of incumbent firms, which can be used to approximate how inframarginal entrants respond to the policy. Concretely, we assume that, absent the reform, the Low- versus High-FR composition of inframarginal entrants would match that observed among pre-reform entrants, and that conditional on this status their behavioral response equals the corresponding incumbent response.¹⁹ For outcomes that are predetermined with respect to the policy, we instead set $\Delta_{AT} = 0$.

6.2 Extensive margin results

We first estimate Equation 34 using the new number of firms as the outcome variable. This recovers the number of new firms due to the policy, from which we derive the share of always-takers and compliers (π_{AT} and π_C , respectively). Columns (1) and (2) in Table 4 show large effects in the number of new firms in exposed industries post-reform, both in levels and in logs.

Columns (3), (4) and (5) estimate the β_k^{DiD} s for Factor R-related outcomes to show that the cohorts of new firms in the exposed industries changed relative to the non-exposed industries in a way that is fully consistent with the policy. New firms in treated industries become 20-24pp more likely to have a Factor R above 28%, which basically doubles the baseline likelihood. The vast majority locate right above the threshold and come from the lowest part of the Factor R distribution. The latter result helps rationalize Figure 1, where a considerable share of bunchers seems to come from the missing mass at zero Factor R. While incumbent bunchers were more likely to come from right below the threshold, the new entry of firms explains the large missing mass at the lower part of the Factor R distribution.

New Entry and Decomposition

¹⁹This is not a theoretical assumption, but rather an empirical assumption. We do not observe how inframarginal new firms would have behaved in the absence of the policy, but we rely on our incumbents' responses as a proxy.

Table 4: Extensive-Margin Difference-in-Differences Estimates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	# New Firms	Log New Firms	Pr(FR > .28)	Pr(.28 < FR < .4)	Pr(FR < .05)	Owner Wages/Revenues	Employees Payroll/Revenues
<i>Pre-period</i>							
2016	-33.184** (14.138)	-0.112 (0.092)	-0.010 (0.011)	-0.008* (0.005)	0.008 (0.014)	0.002 (0.005)	-0.011** (0.005)
<i>Post-period</i>							
2018	150.135*** (36.131)	0.379*** (0.078)	0.209*** (0.014)	0.107*** (0.009)	-0.161*** (0.014)	0.081*** (0.006)	0.000 (0.005)
2019	206.167*** (52.391)	0.452*** (0.069)	0.240*** (0.018)	0.131*** (0.014)	-0.175*** (0.014)	0.093*** (0.007)	-0.006 (0.004)
Mean Dep. Var. Treated	237.29	4.15	0.24	0.06	0.53	0.13	0.05
Mean Dep. Var. Control	266.63	4.05	0.26	0.07	0.49	0.08	0.11
FE by Year	✓	✓	✓	✓	✓	✓	✓
FE by Industry	✓	✓	✓	✓	✓	✓	✓
Reweighted	X	X	✓	✓	✓	✓	✓
Observations	472	472	466	466	466	466	466

Notes: This table reports event-study difference-in-differences estimates of the Factor R reform on extensive-margin outcomes. Each column corresponds to a separate regression. Coefficients are interactions of year indicators with treatment exposure (FR_{act}), with 2017 as the omitted reference year. All specifications absorb year and industry (CNAE) fixed effects. Standard errors clustered at the industry level in parentheses. *Mean Dep. Var.* rows report the 2016 baseline mean of the dependent variable for treated and control groups. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Columns (6) and (7) estimate the β_k^{DiD} s for the different components of Factor R. This is the first step in identifying what motivates the new entry of firms. We find that new firms almost doubled the share of owner wages over revenues, but they do not change employees' payroll over revenues at all. We now turn to tease apart how much of this effect is explained by infra-marginal firms changing their behavior due to the policy, versus the composition of the new firms induced by the policy, following Equation 35.

As explained above, we recover term (I) using the behavioral responses of incumbent firms estimated in Section 5, weighting the estimates by the pre-policy share of new entrants located below and above the 28% threshold. Table XX in the Online Appendix reports the full set of incumbent difference-in-differences coefficients used in this calculation. Table 5 then presents the decomposition of β^{DiD} , together with the resulting selection gap and the comparison between marginal and no-reform entrants across several outcomes.

Table 5: Behavioral and compositional decomposition

	Pr($FR > 0.28$)		Owner W/R		Employee P/R	
	2018	2019	2018	2019	2018	2019
Total effect, β^{DiD}	0.209	0.240	0.081	0.093	0.000	-0.006
Behavioral component, (I)	0.066	0.109	0.019	0.027	-0.001	-0.001
Composition component, (II)	0.143	0.131	0.062	0.066	0.001	-0.005
Composition share	68%	55%	76%	71%	—	—
$E[Y(1) C] - E[Y(1) AT]$	0.453	0.360	0.197	0.181	0.003	-0.014
$E[Y(1) C] - E[Y(0) AT]$	0.519	0.469	0.216	0.208	0.002	-0.015

Notes: Term (I) is the behavioral response of inframarginal entrants, proxied using the incumbent estimates from Section 5. Term (II) is the residual of the total difference-in-differences effect after subtracting (I). The share of marginal entrants is recovered from the difference-in-differences estimate for the number of new firms ($\pi_C = 0.315$ in 2018 and 0.364 in 2019). The selection gap compares marginal and inframarginal entrants under the same post-reform environment (fifth row). The marginal vs. no-reform gap additionally incorporates the behavioral response of inframarginal entrants and compares marginal entrants with the entrants that would have been observed absent the reform (sixth row).

Consistent with the policy incentives, we see that, even when only about a quarter of pre-policy entrants had a Factor R above 28%, the selection gap is very large. Compliers are 45pp more likely to enter with a Factor R above 28% than inframarginal entrants, even after accounting for their behavioral responses. Moreover, marginal entrants are no different from inframarginals in terms of labor intensity. Rather, marginal firms have a 20 pp larger owner wages-to-revenue ratio than inframarginal firms.

What motivates marginal entry? Proposition 1 identifies two mechanisms through which Factor R can raise the value of Simples. Firms may benefit because their unconstrained choices naturally generate a sufficiently high Factor R, which is more likely for sufficiently labor-biased technologies, or because they can manufacture eligibility at low cost. Empirically, we find no evidence that marginal entrants are selected on hired-labor intensity. Instead, their distinguishing Factor R component is owner compensation, consistent with selection on the income-shifting/manufactured-eligibility margin.

Counterfactual

Understanding what marginal entrants would have done in the absence of the reform is central to evaluating its extensive-margin effects. Proposition ?? provides a productivity ordering of these counterfactuals: conditional on the labor-bias parameter, firms induced from *Lucro Presumido* are higher-productivity than firms induced from the outside option. Empirically, we therefore examine whether increased entry into Simples is accompanied

by displacement from alternative formal tax regimes or instead by an increase in firms coming from inactivity/informality.

Table 6 shows the difference-in-differences result for exposed and non-exposed industries regarding new firms by tax regime. The reduction in the number of new firms in Lucro Presumido (column 3) accounts for about 35% of the increase in the number of new firms in the simplified tax regime (column 2). This means that an important part of the new firms in Simples Nacional induced by the policy would have operated in the alternative tax regime.

Table 6: Extensive-Margin: New Firms Entry

	(1) Subject to FR	(2) # Simples Nacional	(3) # Lucro Presumido	(4) # Total
<i>Pre-period</i>				
2016	0.001 (0.006)	-25.244** (10.820)	-3.550 (4.672)	-28.794** (13.138)
<i>Post-period</i>				
2018	0.565*** (0.022)	129.444*** (31.568)	-40.079*** (11.910)	89.365*** (23.927)
2019	0.543*** (0.025)	180.055*** (46.114)	-48.497*** (17.018)	131.558*** (35.782)
Mean Dep. Var. Treated	0.009	206.953	149.814	356.767
Mean Dep. Var. Control	0.000	208.473	13.255	221.727
FE by Year	✓	✓	✓	✓
FE by Industry	✓	✓	✓	✓
Observations	472	564	564	564

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

While the substitution of new entrants in one regime for another is prevalent and straightforward, it is not the only possible counterfactual. Incumbent firms in Lucro Presumido may switch to Simples Nacional (showing up as new entrants), firms close to graduating from Simples Nacional may prefer to not to do so, and existing firms in Simples Nacional may become less likely to close. Although we cannot tease apart these alternative channels, Table A.2 shows the changes in stock resulting from the policy, which summarizes all such channels. Accounting for the small differential trend in the stock of firms in Simples Nacional, Lucro Presumido accounts for 35%-60% of the increase in the stock of firms in Simples Nacional.

Another potential counterfactual is what is usually referred to as *pejotizacao* in Brazil. This is the situation in which an employee is hired through a contractor arrangement,

and wage payments are processed as a business-to-business transaction. This allows the employer to not make contributions on behalf of the employee, and the worker to pay business taxes rather than personal income tax. As the Factor R policy reduced the revenue tax rate, it increased the incentive for employees to switch contract types. We test for this by checking whether the likelihood of coming from an employment contract changed with the policy. Table A.3 shows no compositional effect, suggesting that the marginal new entrant is not more likely to be switching contracts.

7 Revenue Implications

We combine our estimates into a back-of-the-envelope calculation of the revenue implications of the Factor R reform. We decompose the total change in Simples Nacional revenue-tax collection, relative to a no-reform counterfactual, into a mechanical component and the different behavioral responses we have documented. The mechanical term isolates the change in liabilities holding firm behavior fixed at its pre-reform level; the behavioral terms then add, on top of the mechanical baseline, the revenue consequences of each margin the reform activated. We organize the exercise around two populations affected by distinct incentives: incumbent firms, split by whether their pre-policy Factor R was above or below the 28% threshold, and new entrants, split by whether they represent genuinely new activity or a reallocation from other tax regimes. We compute the exercise separately for 2018 and 2019, in each case against that year's no-reform counterfactual.

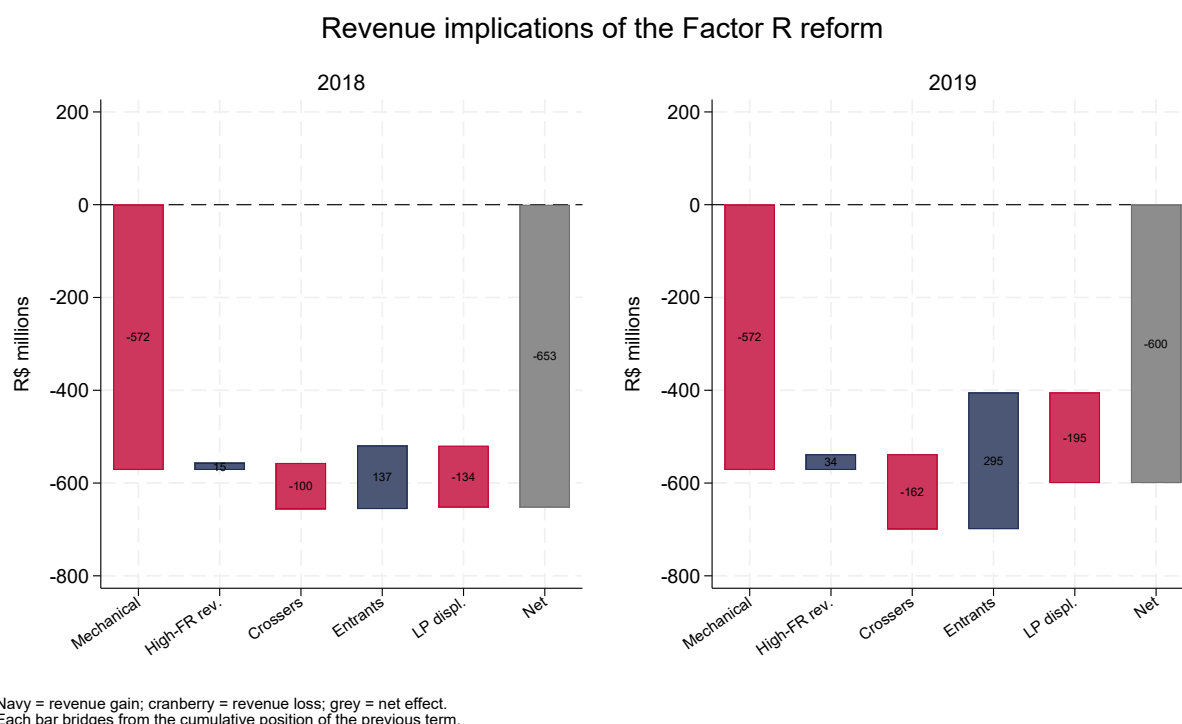
$$\begin{aligned} \Delta \text{Rev}_t = & \underbrace{-(\tau_{VI} - \tau_{III}) R^{hi}}_{(1) \text{ mechanical}} + \underbrace{\tau_{III} \rho_t R^{hi}}_{(2a) \text{ revenue response, high-FR}} - \underbrace{(\tau_V - \tau_{III}) c_t R^{lo}}_{(2b) \text{ crossers, low-FR}} \\ & + \underbrace{\tau_{III} \bar{r}_e N_t^{in}}_{(3a) \text{ pure entrants}} - \underbrace{(\tau_{LP} - \tau_{III}) \bar{r}_e N_t^{LP}}_{(3b) \text{ LP displacement}} \end{aligned} \quad (36)$$

Equation (36) has five terms. The *mechanical* term (1) captures the automatic loss from incumbents with pre-policy Factor R above 28%, who are reassigned from the pre-reform schedule (τ_{VI}) to Anexo III (τ_{III}) without changing behavior; R^{hi} is their pre-reform reported revenue and the term is unambiguously negative. Term (2a) is the behavioral response of these same high-FR incumbents, who increase reported revenues by a fraction ρ_t (our reduced-form revenue response); the additional base is taxed at the new, lower rate τ_{III} , so the term is positive. Term (2b) captures low-FR incumbents who cross the threshold: a fraction c_t move from Anexo V to Anexo III, and since reported revenues are essentially unchanged below the threshold, the effect is a pure rate reduction ($\tau_V - \tau_{III}$) applied to their pre-reform revenue base R^{lo} , yielding a loss. The last two terms are the extensive margin, where $\bar{r}_e$ is an entrant's average revenue and the counts come directly

from the entry regressions (Table 6). Term (3a) is the gain from genuinely new firms (the increase in entrants from inactivity/informality N_t^{in}). Term (3b) is the loss from firms displaced out of Lucro Presumido, N_t^{LP} : because these firms would have paid taxes under LP absent the reform, the relevant cost is only the differential $(\tau_{LP} - \tau_{III})$, not the full Anexo III liability. Incumbent shares and revenue bases are computed separately for the high- and low-FR groups, and entrant counts are cumulated across cohorts so that each year's figure reflects the stock of policy-induced firms then active.

Shared inputs. The exercise uses a stock of $N_{\text{trat}} = 96,000$ treated (Anexo VI) incumbents in the pre-period, split into $s_{hi} = 0.30$ above and $s_{lo} = 0.70$ below the 28% threshold according to their pre-policy Factor R. Group-specific average revenues are $\bar{r}_{hi} = 220,681$ for high-FR and $\bar{r}_{lo} = 308,565$ for low-FR firms, taken from the treated columns of Table 2 (accumulated 12-month). Tax rates are the pre-reform Anexo VI rate $\tau_{VI} = 15\%$, the plain Anexo III rate $\tau_{III} = 6\%$, the payroll-inclusive Anexo III rate $\tau_{III}^P = 9\%$ (applied to firms that report pro-labore to reach or hold above the threshold), the Anexo V rate $\tau_V = 15\%$, and the Lucro Presumido effective rate $\tau_{LP} = 14\%$. On the extensive margin, effects are scaled by the number of treated CNAEs, $0.6 \times 86 \approx 51.6$.

Figure 5: Revenue Decomposition



Notes: Back-of-the-envelope decomposition of the change in Simples Nacional revenue-tax collection relative to a no-reform counterfactual, computed separately for 2018 and 2019 following Equation 36. Red bars represent negative numbers, blue bars represent positive numbers. Each bar builds on the previous one. The gray bar represents the total effect. All figures in millions of R\$ per year. The mechanical term and the high-FR revenue response are evaluated at the plain Anexo III rate (6%), as these margins involve no additional payroll reporting; the crossing, entry, and displacement terms are evaluated at the payroll-inclusive effective rate (9%), since these firms report pro-labore to reach or remain above the 28% threshold. See text for sources of each component.

8 Conclusions

This paper studies how firms respond and select into preferential policies when benefits depend on characteristics that firms can themselves adjust. Brazil's Factor R reform provides unusual variation to study both margins. The reform sharply reduced revenue tax rates for firms with payroll above 28 percent of revenues, increasing the value of the simplified tax regime both for firms that were naturally labor intensive and for firms that could reach the threshold by changing their behavior.

The evidence reveals that large tax incentives need not generate large responses along the real margin they favor. Incumbent firms that were already above the threshold received the tax cut automatically. Despite a substantial increase in reported revenues, their employee payroll did not increase. Firms initially below the threshold faced the same potential tax cut but had to raise their payroll-to-revenue ratio to obtain it. They responded almost entirely by shifting owner income from dividends to wages, rather than by in-

creasing employee payroll or reducing reported revenue. Thus, the limited employment response reflects more than the presence of an easily exploited eligibility rule: even firms that did not need to manipulate eligibility showed no detectable increase in employment.

The extensive margin reinforces this conclusion. The reform increased entry into Simples Nacional by about 50 percent, but the marginal entrants were not more intensive in hired labor. Instead, they were disproportionately firms that used owner compensation to satisfy Factor R. Moreover, around 35 percent of the additional flow into Simples displaced firms that would otherwise have entered another formal tax regime. Large take-up therefore need not indicate a correspondingly large effect on the economic activity a policy is designed to favor.

More broadly, our results highlight two forces that matter for the design of preferential policies. The real behavior policymakers seek to affect may itself be relatively inelastic, and alternative ways of qualifying for a benefit may be considerably cheaper. Heterogeneity in these adjustment costs then affects not only the behavior of existing beneficiaries but also the composition of those induced to participate. Evaluating such policies therefore requires looking beyond take-up and statutory eligibility measures to understand both how firms obtain preferential treatment and who responds to the policy.

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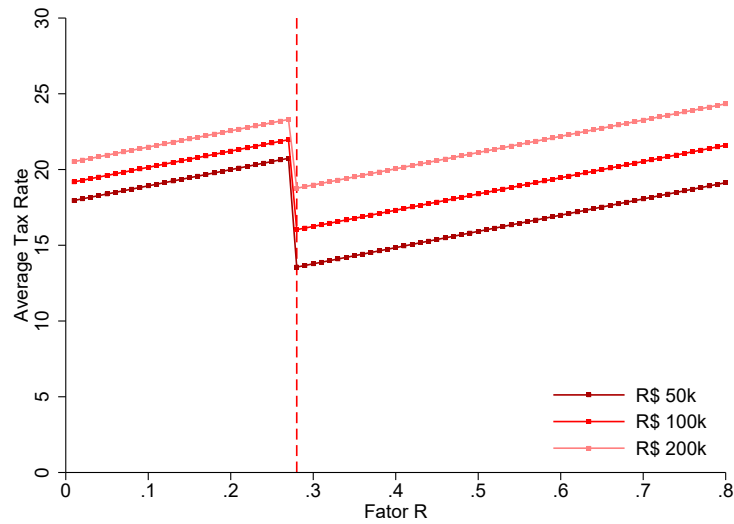
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A Appendix A: Additional Figures and Tables

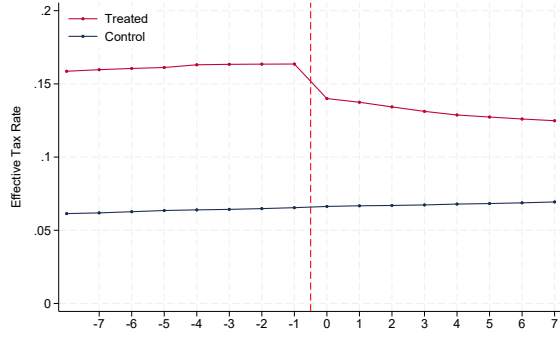
A.1 Figures

Figure A.1: Change in Effective Tax Rate including All Wage Taxes

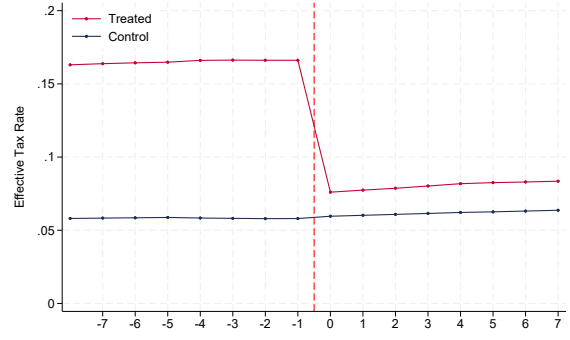


Notes: This Figure plots the average tax rate (average revenue tax + INSS) across the Factor R distribution, for different revenue levels.

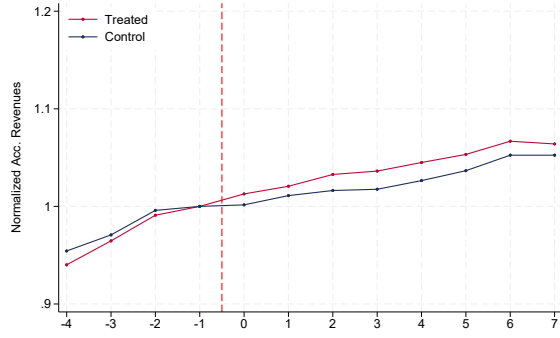
Figure A.2: Treatment vs. Control: Raw trends for incumbent firms



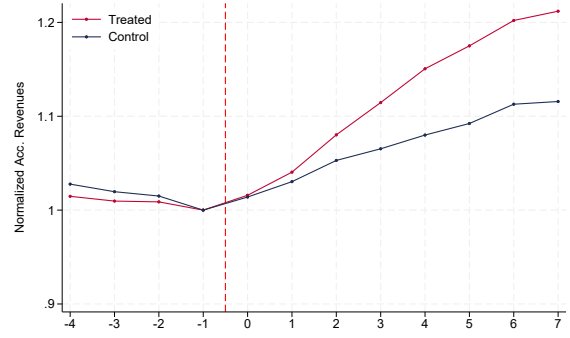
(a) Av. Revenue Tax Rate - Low FR



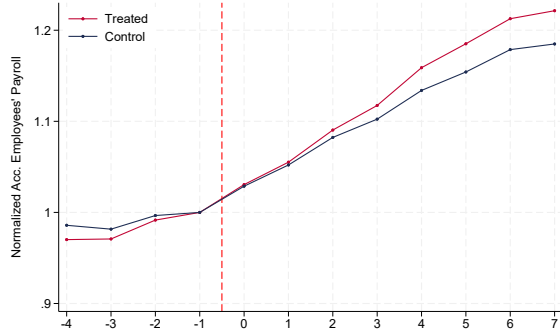
(b) Av. Revenue Tax Rate - High FR



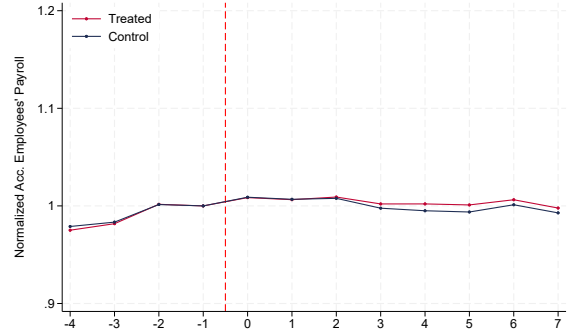
(c) Annual Revenues - Low FR



(d) Annual Revenues - High FR



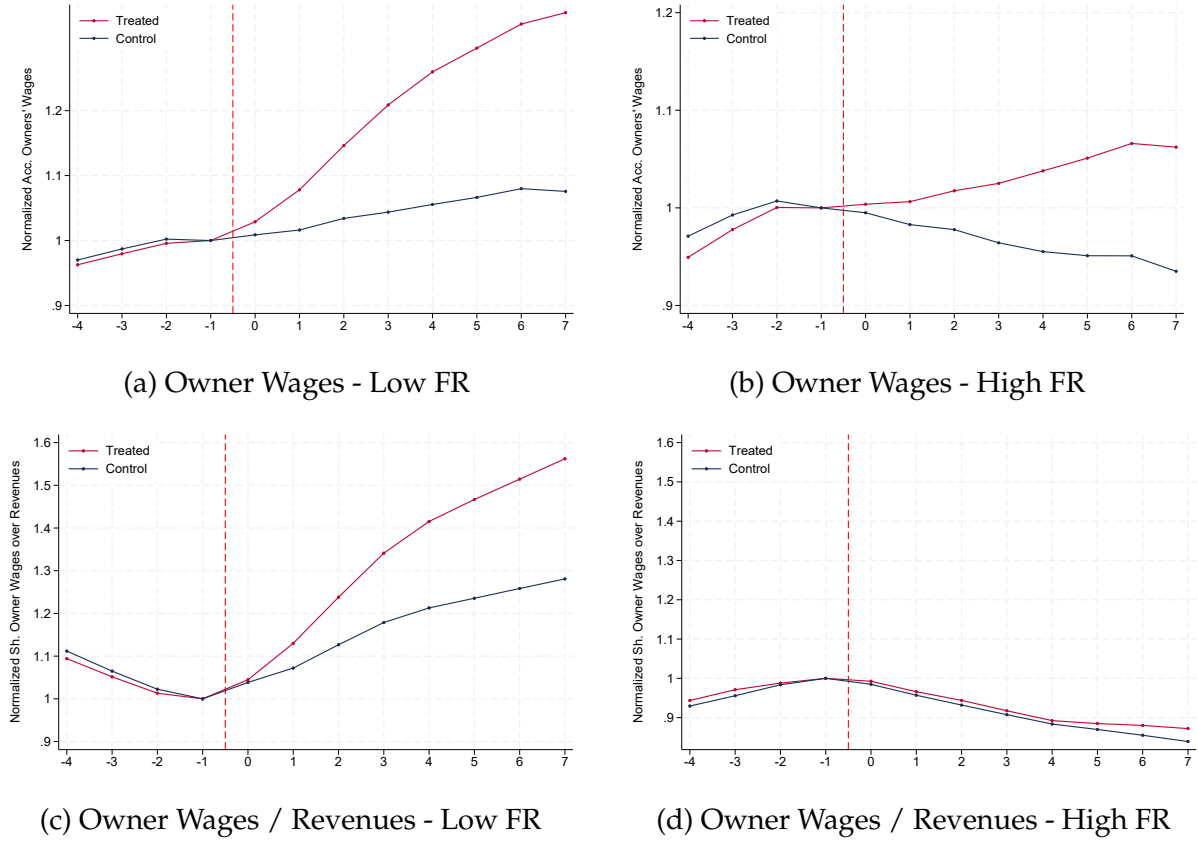
(e) Employees' Payroll - Low FR



(f) Employees' Payroll - High FR

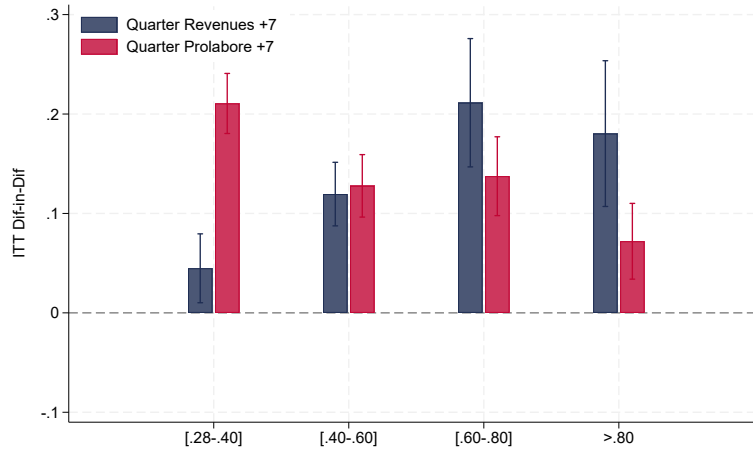
Notes: This Figure shows the raw trends comparing treated vs control incumbent firms. Panels on the left correspond to Low FR firms, while panels on the right correspond to High Factor R firms. Panel (a) and (b) plot the evolution of the average revenue tax rate. Panel (c) and (d) plot the evolution of the average revenues, normalized in the last quarter of 2017. Panel (e) and (f) plot the evolution of the average employees' payroll, normalized in the last quarter of 2017. The vertical dashed bar indicates the introduction of the reform. Revenues and employees' payroll were previously expressed in real terms.

Figure A.3: Treatment vs. Control: Raw trends for incumbent firms (more outcomes)



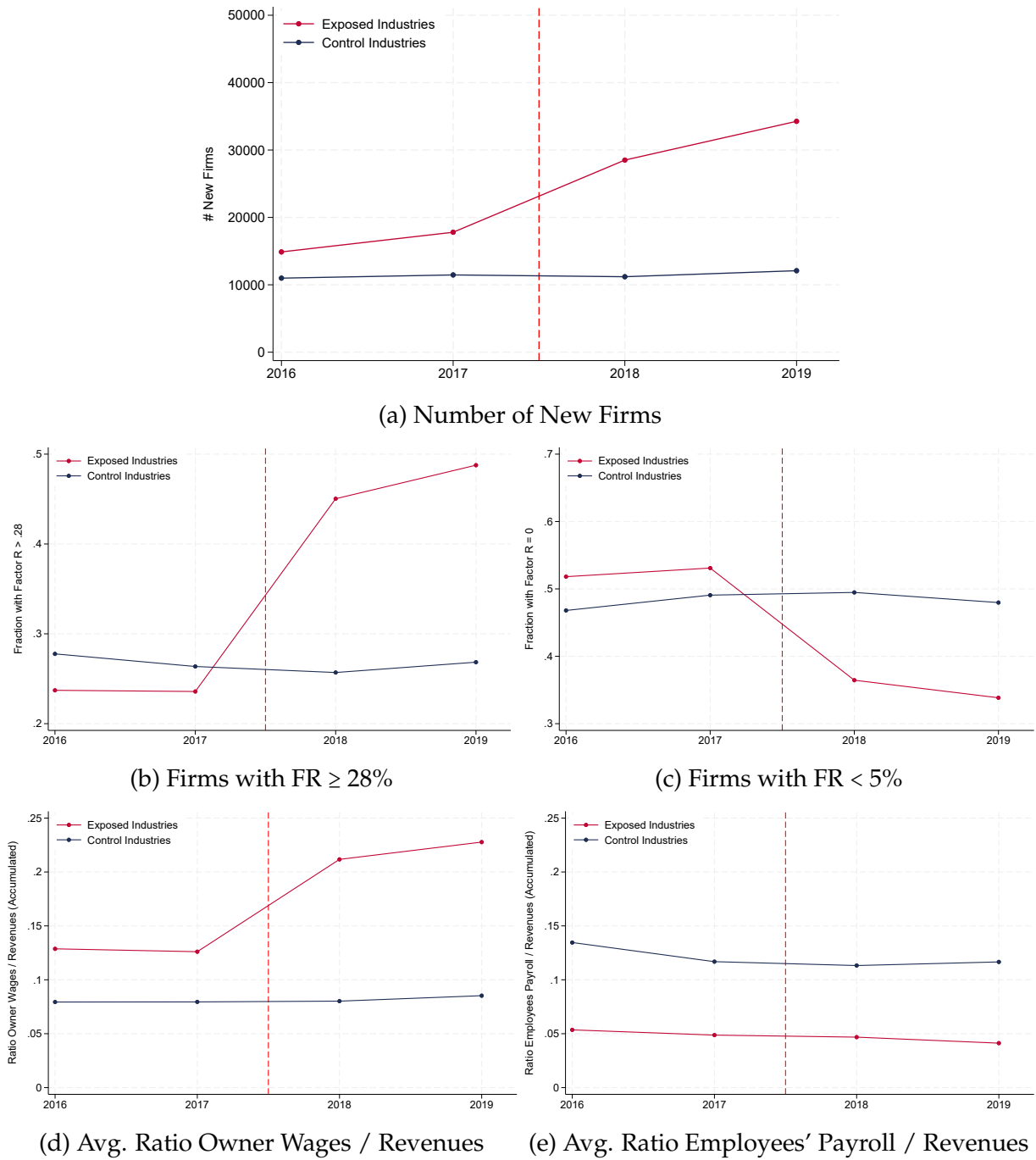
Notes: This Figure shows the raw trends comparing treated vs control incumbent firms. Panels on the left correspond to Low FR firms, while panels on the right correspond to High Factor R firms. Panel (a) and (b) plot the evolution of the average wages paid to the firm's owners, normalized in the last quarter of 2017. Panel (c) and (d) plot the evolution of the share of revenues used to pay owner wages, normalized in the last quarter of 2017. The vertical dashed bar indicates the introduction of the reform. Revenues and owner wages were previously expressed in real terms

Figure A.4: ITT for Incumbent Firms based on pre-policy Factor R

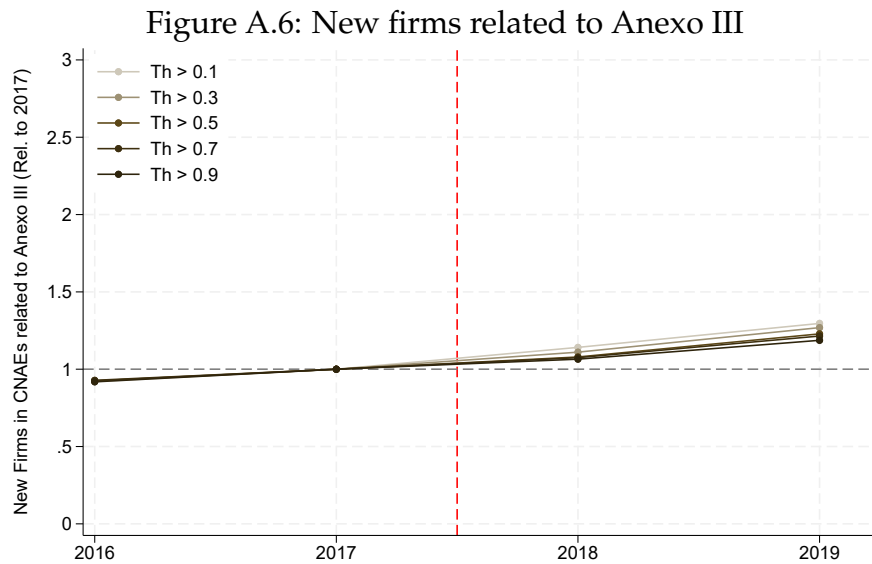


Notes: This Figure shows the β_7 from Equation 33 for different samples based on the pre-policy location in the Factor R distribution. Blue bars use real revenues as outcome, and red bars use owner wages. From left to right, the sample moves further away from the 28% threshold.

Figure A.5: Raw trends for new firms (exposed vs. non-exposed industries)



Notes: This Figure shows the raw trends comparing treated vs control industries. The unit of observation is the industry-year, and it is constructed using information on new firms within each industry-year. Panels on the left correspond to Low FR firms, while panels on the right correspond to High Factor R firms. Panel (a) plots the evolution of the average number of new firms. The remaining plots show the evolution of the new firms' composition. Panel (b) plots the share of new firms that entered with a Factor R above 28%. Panel (c) plots the share of new firms that entered with a Factor R below 5%. Panel (d) and (e) decompose the Factor R by each component of the numerator: owner wages and employees' payroll, respectively. The vertical dashed bar indicates the introduction of the reform. Revenues, owner wages and employees' payroll were previously expressed in real terms. Averages across industries are weighted by each industry's relative size, defined as the number of new firms in the baseline year.



Notes: This Figure shows the evolution in the number of new firms for the economic activities related to Anexo III. Each line represents a different threshold to define the group of industry codes. For example, the lightest uses the industry codes for which at least 10% of the firms pre-policy were listed in Anexo III. The darkest only considers those industry codes for which at least 90% of the firms were listed in Anexo III. The red dashed line indicates the policy event.

A.2 Tables

Table A.1: Broad economic activities by Anexo (Simples Nacional)

Control: Anexo IV	Treated: Anexo VI
Construction (incl. subcontracting) and interior/landscaping services	Health professionals (medical, dental, veterinary; psychology and related clinics)
Security, cleaning, and maintenance services	Regulated professional services (architecture/engineering; technical consulting and auditing)
Legal services (advocacy)	Other intellectual services (e.g. advertising/journalism; translation/interpreting; appraisals)

Notes: This table provides illustrative examples of activities classified under Anexo IV (control) and Anexo VI (treated) prior to the reform. Categories are based on LC 123/2006, art. 18, §5°-C and §5°-I, and are shown for exposition rather than as an exhaustive list.

Table A.2: Extensive-Margin: Stock of Firms

	(1)	(2)	(3)	(4)
	Subject to FR	# Simples Nacional	# Lucro Presumido	# Total
<i>Pre-period</i>				
2016	-0.001 (0.001)	-141.914 (158.246)	14.671 (51.056)	-127.243 (173.791)
<i>Post-period</i>				
2018	0.534*** (0.022)	466.549** (187.235)	-207.034*** (53.573)	259.515 (170.653)
2019	0.526*** (0.022)	839.358** (362.791)	-303.281*** (84.235)	536.077 (346.132)
Mean Dep. Var. Treated	0.014	1724.233	2512.628	4236.860
Mean Dep. Var. Control	0.001	2466.218	637.964	3104.182
FE by Year	✓	✓	✓	✓
FE by Industry	✓	✓	✓	✓
Observations	558	564	564	564

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Extensive-Margin Difference-in-Differences Estimates

	(1)	(2)	(3)	(4)	(5)
	# New Firms	Log New Firms	Sh. Employees t-1	Sh. Priv. Employees t-1	Sh. Top 5% Employees t-1
<i>Pre-period</i>					
2016	-33.184** (14.138)	-0.112 (0.092)	0.005 (0.020)	-0.011 (0.016)	0.020*** (0.006)
<i>Post-period</i>					
2018	150.135*** (36.131)	0.379*** (0.078)	-0.001 (0.008)	-0.004 (0.011)	-0.005 (0.006)
2019	206.167*** (52.391)	0.452*** (0.069)	0.001 (0.008)	0.003 (0.011)	-0.006 (0.007)
Mean Dep. Var. Treated	237.29	4.15	0.47	0.32	0.16
Mean Dep. Var. Control	266.63	4.05	0.37	0.28	0.07
FE by Year	✓	✓	✓	✓	✓
FE by Industry	✓	✓	✓	✓	✓
Rewighted	X	X	✓	✓	✓
Observations	472	472	466	466	466

Notes: This table reports event-study difference-in-differences estimates of the Fator R reform on extensive-margin outcomes. Each column corresponds to a separate regression. Coefficients are interactions of year indicators with treatment exposure (FR_{act}), with 2016 as the omitted reference year. All specifications absorb year and industry (CNAE) fixed effects. Standard errors clustered at the industry level in parentheses. *Mean Dep. Var.* rows report the 2016 baseline mean of the dependent variable for treated and control groups.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Anexo VI tax rates: pre- vs post-reform (Factor R)

Pre-reform		Post-reform		
Annual Revenues (R\$)	Tax Rate	Annual Revenues (R\$)	Tax Rate	
			FR < .28	FR ≥ .28
180,000	16.93%	180,000	15.50%	6.00%
180,000–360,000	17.72%	180,000–360,000	18.00%	11.20%
360,000–540,000	18.43%	360,000–720,000	19.50%	13.50%
540,000–720,000	18.77%	720,000–1,800,000	20.50%	16.00%
720,000–900,000	19.04%	1,800,000–3,600,000	23.00%	21.00%
900,000–1,080,000	19.94%	3,600,000–4,800,000	30.50%	33.00%
1,080,000–1,260,000	20.34%			
1,260,000–1,440,000	20.66%			
...	...			
3,420,000–3,600,000	22.45%			

B Revenue Elasticity

Figure 4b shows that High-FR firms increase the reported revenues in response to the automatic eligibility for the revenue tax cut. To translate the reduced form result into a revenue elasticity, we need to account for the fact that the revenue tax rate depends on the revenue bracket, which may change due to the reform. We do so by instrumenting the revenue tax rate with treatment status and performing 2SLS on the log of revenues. Note that we are forced to use the log of revenues rather than a Poisson specification to avoid the *forbidden regression* in 2SLS. However, as our sample of incumbent firms comprises firms with positive revenues in all periods, we are not affecting the sample and the reduced form estimates are virtually equivalent for the particular outcome of revenues.

We define the elasticity to the net-of-tax rate between t and $t - 1$ as:

$$\varepsilon_{R,1-\tau} = \frac{\mathbb{E} \left[\log \left(\frac{R_{i,t}}{R_{i,-1}} \right) \mid \text{Treated} = 1 \right] - \mathbb{E} \left[\log \left(\frac{R_{i,t}}{R_{i,-1}} \right) \mid \text{Treated} = 0 \right]}{\mathbb{E} \left[\log \left(\frac{1-\tau_{i,t}}{1-\tau_{i,-1}} \right) \mid \text{Treated} = 1 \right] - \mathbb{E} \left[\log \left(\frac{1-\tau_{i,t}}{1-\tau_{i,-1}} \right) \mid \text{Treated} = 0 \right]} \quad (37)$$

Where the choice of t recovers different time-horizon elasticities. The numerator captures the firm's response, and the denominator measures the size of the incentives. Equation 38 shows the 2SLS equations that we estimate to recover Equation 37. The underlying identifying assumption is that the treated and control groups would follow parallel trends in the absence of the policy.

$$\begin{aligned}\Delta_{-1,t} \ln y_i &= \alpha_0 + \varepsilon_{R,t} \Delta_{-1,t} \ln(1 - \tau_i) + u_{it} \\ \Delta_{-1,t} \ln(1 - \tau_i) &= \gamma_0 + \pi \mathbf{1}[\text{Anexo VI} = 1] + v_{it}\end{aligned}\tag{38}$$

We find sizable revenue elasticities with respect to the revenue tax rate. Table B.5 shows that two years after the policy, High FR firms have increased reported revenues by 1% for each 1% increase in the net-of-tax rate. Moreover, consistent with the previous patterns, firms seem to be gradually responding as the coefficients grow over time.

Table B.5: Revenue Elasticity Estimates Over Time Horizons (High FR)

	12-month Revenue		Quarter Revenue	
	1 Year	2 Year	1 Year	2 Year
ε_R	.392	.887	.994	1.374
SE	(0.045)	(0.063)	(0.067)	(0.084)
CI	[0.30, 0.48]	[0.76, 1.01]	[0.86, 1.13]	[1.21, 1.54]
Reduced Form	.040	.089	0.101	0.137
First Stage (γ_1)	0.101	0.100	0.101	0.100
Observations	34,014	34,014	34,014	34,014

Notes: This table shows the coefficients estimated from Equation 38 for different time horizons. Columns 1 and 2 use as outcome the log of 12-month accumulated revenues (Factor R denominator). Columns 3 and 4 use the monthly average of the last quarter of 2018 and 2019, respectively. The sample contains only firms in the High FR group.

C Details for Revenue Implications

This section provides the details behind the revenue calculations reported in Section 7. We focus on the medium-term effect of the reform and evaluate behavioral responses using the last quarter of 2019, the latest period in our event study. This is useful because the behavioral responses build gradually following the reform.

One quarter before the policy's implementation, there were about $N^{treat} = 96,000$ existing firms in activities affected by the reform. Based on their pre-reform Factor R, 30% belong to the High-FR group and 70% to the Low-FR group. Average 12-month reported revenues in 2017Q4 are R\$220,681 among High-FR firms and R\$308,565 among Low-FR firms, as reported in Table 2.

Mechanical Effect

The mechanical effect holds firms' reported revenues and all other behavior fixed at their pre-reform levels. High-FR firms were already above the 28 percent threshold and therefore became automatically eligible for Anexo III after the reform. The mechanical change in revenue collection is consequently:

$$\Delta Rev^{mech} = -(\tau_{VI} - \tau_{III})R^{hi}.$$

Where $(\tau_{VI} - \tau_{III})$ is the difference in the tax rates induced by the reform. R^{hi} is the total revenue of firms mechanically affected by the reform which equals the share of firms above 28% pre-reform times their average revenues.¹ Using the values above, $\Delta Rev^{mech} = -R\$ 572$ millions.

Incumbent Firms: reported-revenue increase of High-FR firms

High-FR firms subsequently increase reported revenues in response to the lower revenue tax rate. For the medium-term calculation, we use the reduced-form effect in 2019Q4, the last period in our event study. Table 3 reports an estimated increase in quarterly reported revenue of $\hat{\rho}_{2019Q4} = .137$. This differs from the accumulated 12-month revenue because the latter averages together earlier quarters in which firms had not yet fully adjusted. Therefore, $\hat{\rho}_{2019Q4}$ better captures the revenue response prevailing by the end of our estimation period.

As a result, $\Delta Rev^R = \tau_{III} \times \hat{\rho}_{2019Q4} \times R^{hi} = R\$ 52$ millions denotes the increase in revenue collection resulting from the incumbent firms' revenue response.

Incumbent Firms: Income-shifting response of Low-FR firms

Low-FR firms are not automatically eligible for the tax reduction. The relevant revenue effect therefore depends on both how many firms are induced to cross the 28% threshold and the pre-reform revenue of those firms. Because the conceptual framework predicts that the profitability of manufacturing eligibility may vary systematically with firm scale, we do not assume that induced crossers have the average revenue of all Low-FR incumbents. Instead, we estimate these objects directly using the Low-FR incumbent sample.

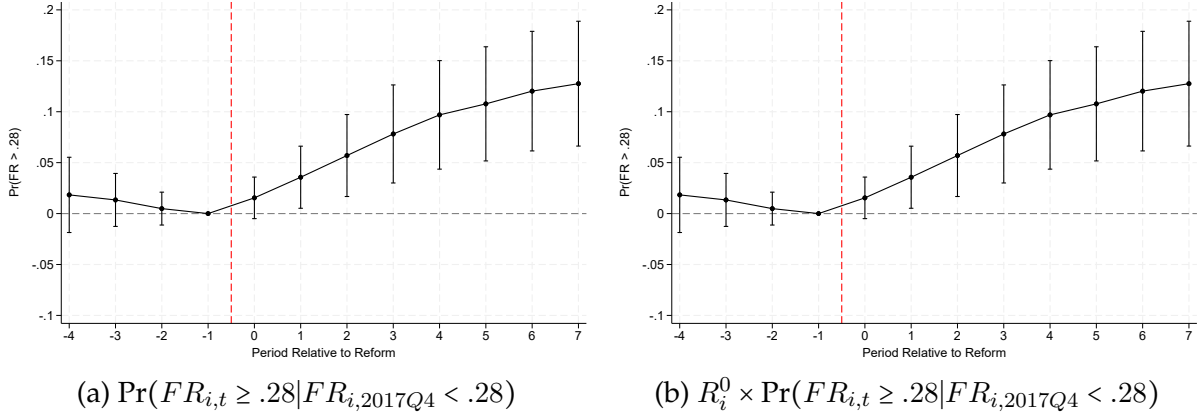
Let $\hat{c}$ denote the fraction of initially Low-FR firms induced by the reform to locate above the eligibility threshold. We can estimate $\hat{c}$ using $\Pr(FR_{i,2019Q4} \geq .28 | FR_{i,2017Q4} < .28)$ as the outcome variable in Equation 33.

Second, to recover the revenue base of the firms induced to cross, let R_i^0 denote the firm's pre-reform 12-month revenue in 2017Q4, and estimate the same specification using $Y_{i,2019Q4}^{R,C} = R_i^0 \times \Pr(FR_{i,2019Q4} \geq .28 | FR_{i,2017Q4} < .28)$. Let's call $\hat{m}$ the resulting coefficient. Then $\hat{m}$ measures the pre-reform revenue mass of induced crossers per initially Low-FR

¹ $R^{hi} = 96,000 \times 0.30 \times 220,681 = R\6.356 billions.

firm.

Figure C.7: Treatment vs. Control (Low Factor R pre-reform)



Notes: This Figure shows

Therefore, the change in revenue collection due to income-shifting can be expressed as:

$$\Delta Rev^{cross} = -(\tau_V - \tau_{III}^P) N^{lo} \hat{m},$$

where τ_{III}^P takes into account that crossing is achieved through income-shifting, implying payroll tax payments on 28% of the reported revenues.²

Extensive Margin

To account for the extensive margin in the revenue calculation, we use the reform's effect on the stock of firms, rather than only on new entrants. While the new entrants analysis cleanly illustrates selection into *Simple Nacional*, it misses several extensive-margin responses that matter for revenue. First, new firms that register in year t , but only made positive revenues in $t + 1$ do not show up in the definition of new entrants used in Table 4. Second, existing firms in *Lucro Presumido* may decide to switch to *Simple Nacional*, affecting tax payments and the stock of firms in both regimes. Third, firms in *Simple Nacional* that would have graduated to *Lucro Presumido* may decide not to do so. Finally, firms in *Simple Nacional* may become less likely to close, affecting the exit margin.

The stock estimates in Table A.2, therefore, summarize the cumulative effect of these margins on the number of firms operating under each tax regime at a given point in time. For the medium-term revenue calculation, we use the 2019 estimates. Since the unit of observation in these regressions is the CNAE-year, the coefficients measure the change in

²As most crossers were already reporting some owner wages, adding payroll taxes for the entire 28% of revenues leads to an overestimate of the counterfactual paymetns and, therefore, a lower bound of the impact in government's budget.

the number of firms per exposed CNAE relative to non-exposed CNAEs. We aggregate these reduced-form estimates across the 86 CNAEs classified as exposed to the reform.

Counterfactual: no registration

We first consider the part of the increase in the stock of firms in *Simples Nacional* whose counterfactual is not revenue registration. To isolate this margin, we use the combined stock of firms in *Simples Nacional* and *Lucro Presumido*, reported in column (4) of Table A.2. Reallocation between the two tax regimes has no effect on this outcome: a firm moving from *Lucro Presumido* to *Simples Nacional*, for example, increases the stock of the latter by one and reduces the stock of the former by one. The effect on their combined stock therefore isolates the net increase in firms operating in these two formal tax regimes.³

In 2019, the estimated effect on the combined stock is

$$\widehat{\beta}_{2019}^{Total} = 536.077$$

firms per exposed CNAE. We therefore estimate the number of additional registered firms whose counterfactual is the outside option as

$$\widehat{N}^0 = N^{CNAE} \widehat{\beta}_{2019}^{Total} = 86 \times 536.077 = 46,103.$$

This object incorporates not only additional entry, but also any policy-induced reduction in exit that increases the stock of firms in the two regimes. Let $\bar{R}^0$ denote the average annual reported revenue used to value these additional firms. Since these firms operate in *Simples Nacional* under the reform and would not have generated business-tax revenue in the counterfactual, their contribution to government revenue is

$$\Delta Rev^0 = \tau_{III}^P \bar{R}^0 \widehat{N}^0,$$

where τ_{III}^P is the effective tax rate under the favorable Factor R schedule, including the tax cost associated with the owner-wage reporting used to satisfy the eligibility requirement.

Counterfactual: Lucro Presumido tax regime

We next consider firms that operate in *Simples Nacional* because of the reform but whose counterfactual is *Lucro Presumido*. Column (3) of Table A.2 directly captures this displacement margin. In 2019, the reform reduces the stock of firms in *Lucro Presumido* by

$$\widehat{\beta}_{2019}^{LP} = -303.281$$

³Table A.3 and Figure A.6 rule out alternative counterfactual.

firms per exposed CNAE. Thus, the estimated number of firms displaced from *Lucro Presumido* is

$$\widehat{N}^{LP} = -N^{CNAE} \widehat{\beta}_{2019}^{LP} = 86 \times 303.281 = 26,082.$$

Notice that the two counterfactuals exactly account for the estimated increase in the stock of firms in *Simples Nacional*:

$$839.358 = 536.077 + 303.281.$$

The first component represents an increase in the combined stock of formal firms, whereas the second represents reallocation from *Lucro Presumido* to *Simples Nacional*.

For firms displaced from *Lucro Presumido*, the fiscal cost is not the full tax liability paid under *Simples Nacional*, since these firms would have paid taxes under *Lucro Presumido* in the absence of the reform. Let $\bar{R}^{LP}$ denote their average annual reported revenue. Their contribution to the revenue effect is therefore

$$\Delta Rev^{LP} = -(\tau_{LP} - \tau_{III}^P) \bar{R}^{LP} \widehat{N}^{LP}.$$

Thus, this component captures only the change in tax payments generated by reallocating a firm from *Lucro Presumido* to the favorable *Simples Nacional* schedule.

Robustness: Honest pre-trend exercise

The stock specification exhibits some evidence of differential pre-treatment movements, although the pre-reform coefficients are statistically indistinguishable from zero. In particular, relative to the omitted year 2017, the 2016 coefficient is -127.243 firms for the combined stock of *Simples Nacional* and *Lucro Presumido*, and 14.671 firms for the stock in *Lucro Presumido*. Rather than interpreting the lack of statistical significance as evidence that the pre-treatment differential trend is exactly zero, we assess the sensitivity of the 2019 estimates using the approach of [Rambachan and Roth, 2023](#).

We implement their relative-magnitude restriction, which allows violations of parallel trends after the reform to be bounded relative to the magnitude of the differential trend observed before the reform. Let $\bar{M}$ denote the maximum post-reform violation relative to the pre-reform benchmark. We report robust confidence intervals for the 2019 coefficients for $\bar{M} = \{0.5, 1, 1.5, 2\}$, together with the breakdown value of $\bar{M}$ at which the robust confidence interval first includes zero. For example, $\bar{M} = 1$ allows post-reform deviations from parallel trends to be as large as the deviation observed in the pre-reform period.

Because our event study contains only one non-normalized pre-reform coefficient, this exercise should be interpreted as a sensitivity analysis rather than as an estimate of the

counterfactual trend. Our baseline revenue calculation continues to use the reduced-form 2019 coefficients in Table [A.2](#); the HonestDiD exercise quantifies how sensitive the implied extensive-margin effects are to departures from exact parallel trends.

How do Firms Behave in Simplified Tax Regimes?

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Online Appendix For Publication

D Conceptual Framework: General Results and Proofs

This appendix provides the formal derivations and proofs for the conceptual framework in Section 4. The analysis remains general with respect to the firm's production technology and the functional forms governing input costs, revenue reporting, and owner compensation.

Throughout, we suppress the firm subscript i whenever this does not create ambiguity. A firm's non-scale technology is denoted by F , and its productivity shifter by A , so true revenue is

$$R = AF(L, e). \quad (39)$$

When studying comparative statics in A , the non-scale technology F is held fixed. All input, compensation, and reporting choices are nevertheless allowed to re-optimize with A .

Let

$$x = (e, L, \tilde{R}, w^o)$$

denote the firm's vector of choices and

$$r(x) = \frac{wL + w^o}{\tilde{R}}$$

its Factor R . We restrict attention to operating allocations with positive reported revenue.

Conditional on schedule $j \in \{H, L\}$, the owner's payoff is

$$u_j(x; A, F) = AF(L, e) - \mathcal{T}_j(\tilde{R}) - wL - v(e) - \Phi(L) - c(AF(L, e) - \tilde{R}) - \Gamma(w^o). \quad (40)$$

The favorable schedule satisfies

$$\mathcal{T}_L(\tilde{R}) < \mathcal{T}_H(\tilde{R}) \quad \text{for all } \tilde{R} > 0 \quad (41)$$

over the relevant revenue support.

The maximized operating values introduced in the main text are

$$\begin{aligned} \Pi_H(A; F) &= \max_x u_H(x; A, F), \\ \Pi_L(A; F) &= \max_x u_L(x; A, F), \\ \Pi_T(A; F) &= \max_{x: r(x) \geq T} u_L(x; A, F). \end{aligned} \quad (42)$$

Let x^L denote an unconstrained low-tax optimum and define

$$r^N(A; F) = r(x^L). \quad (43)$$

As in the main text, if the low-tax problem admits multiple optimal allocations, we assume that they generate the same employee-payroll and owner-wage shares. Hence p^N , ω^N , and r^N are uniquely defined even when x^L itself is not unique.

When the high-tax problem has multiple optima, statements below that a non-responding firm retains its pre-reform allocation should be interpreted using a fixed selection rule across policy states. None of the value comparisons depends on this convention.

D.1 Operating values and the Factor R gain

Recall

$$B(A; F) = \Pi_L(A; F) - \Pi_H(A; F), \quad C(A; F) = \Pi_L(A; F) - \Pi_T(A; F). \quad (44)$$

The first object is the unconstrained value of receiving the favorable tax schedule. The second is the optimized loss generated by imposing the Factor R eligibility condition while holding the favorable schedule fixed.

Lemma D.1 (Basic properties of B and C). *Suppose the problems in (42) admit solutions and the high-tax optimum has positive reported revenue. Then*

$$B(A; F) > 0, \quad C(A; F) \geq 0.$$

Moreover, if $r^N(A; F) \geq T$, then $C(A; F) = 0$.

Proof. Let x^H attain $\Pi_H(A; F)$. By schedule dominance,

$$u_L(x^H; A, F) > u_H(x^H; A, F) = \Pi_H(A; F).$$

Since Π_L maximizes u_L over the unrestricted choice set,

$$\Pi_L(A; F) \geq u_L(x^H; A, F) > \Pi_H(A; F),$$

and therefore $B(A; F) > 0$.

The feasible set defining Π_T is a subset of the feasible set defining Π_L , implying

$$\Pi_T(A; F) \leq \Pi_L(A; F),$$

and hence $C(A; F) \geq 0$.

Finally, suppose $r^N(A; F) \geq T$. The unconstrained low-tax optimum x^L is then feasible for the problem defining Π_T . Thus

$$\Pi_T(A; F) \geq u_L(x^L; A, F) = \Pi_L(A; F).$$

Together with $\Pi_T \leq \Pi_L$, this implies $\Pi_T = \Pi_L$ and therefore $C = 0$. $\square$

Lemma D.2 (Value generated by Factor R). *The increase in maximized operating value generated by Factor R is*

$$G(A; F) = [B(A; F) - C(A; F)]_+. \quad (45)$$

Proof. After the reform, the high schedule applies to non-qualifying allocations. Define

$$\Pi_{H, < T}(A; F) \equiv \sup_{x: r(x) < T} u_H(x; A, F).$$

The firm's actual post-reform operating value is therefore

$$\Pi^1(A; F) = \max\{\Pi_{H, < T}(A; F), \Pi_T(A; F)\}.$$

We first establish

$$\Pi^1(A; F) = \max\{\Pi_H(A; F), \Pi_T(A; F)\}. \quad (46)$$

Because the non-qualifying choice set is a subset of the unrestricted high-schedule choice set,

$$\Pi_{H, < T} \leq \Pi_H.$$

Hence

$$\Pi^1 \leq \max\{\Pi_H, \Pi_T\}.$$

For the reverse inequality, consider a high-tax optimum x^H . If $r(x^H) < T$, then x^H remains feasible under the high schedule after the reform and therefore

$$\Pi_{H, < T} = \Pi_H.$$

If instead $r(x^H) \geq T$, then x^H is feasible for the qualifying low-tax problem and schedule dominance implies

$$\Pi_T \geq u_L(x^H) > u_H(x^H) = \Pi_H.$$

Thus, in either case, $\Pi^1 \geq \Pi_H$. Since $\Pi^1 \geq \Pi_T$ by construction, (46) follows.

The gain relative to the pre-reform operating value is consequently

$$G = \max\{\Pi_H, \Pi_T\} - \Pi_H = \max\{0, \Pi_T - \Pi_H\}.$$

Using

$$\Pi_T - \Pi_H = (\Pi_L - \Pi_H) - (\Pi_L - \Pi_T) = B - C$$

gives (45). □

D.1.1 Shadow-cost interpretation of C

The definition of C does not require differentiability. Under additional regularity, however, it admits a useful shadow-cost representation.

For an arbitrary eligibility threshold t , define

$$\Pi(t; A, F) = \max_{x: wL + w^o - t\tilde{R} \geq 0} u_L(x; A, F), \quad C(t; A, F) = \Pi_L(A; F) - \Pi(t; A, F). \quad (47)$$

Thus $C(T; A, F) = C(A; F)$.

Suppose that over an interval of thresholds the constrained optimum is differentiable, the eligibility constraint binds, and the usual constraint qualification holds. Let $\lambda(t) \geq 0$ denote the multiplier on

$$wL + w^o - t\tilde{R} \geq 0.$$

The envelope theorem yields

$$\frac{\partial \Pi(t; A, F)}{\partial t} = -\lambda(t)\tilde{R}(t),$$

and therefore

$$\frac{\partial C(t; A, F)}{\partial t} = \lambda(t)\tilde{R}(t) \geq 0. \quad (48)$$

If $r^N < T$ and these regularity conditions hold along the path from r^N to T ,

$$C(A; F) = \int_{r^N(A; F)}^T \lambda(t)\tilde{R}(t) dt. \quad (49)$$

Thus C can be interpreted as the cumulative shadow cost of tightening the eligibility requirement from the firm's preferred low-tax Factor R to the statutory threshold.

D.2 Natural eligibility and labor intensity

At the unconstrained low-tax allocation define

$$p^N(A; F) = \frac{wL^L(A; F)}{\tilde{R}^L(A; F)}, \quad \omega^N(A; F) = \frac{w^{o,L}(A; F)}{\tilde{R}^L(A; F)}. \quad (50)$$

Then

$$r^N(A; F) = p^N(A; F) + \omega^N(A; F). \quad (51)$$

These are equilibrium objects. The general framework does not impose a scalar ordering on the non-scale technologies F . Different technologies may therefore generate different natural employee-payroll intensities, owner-compensation shares, and natural Factor R values.

Natural eligibility is characterized directly by

$$r^N(A; F) \geq T.$$

By Lemma D.1, this condition implies

$$C(A; F) = 0,$$

and Lemma D.2 then gives

$$G(A; F) = B(A; F) > 0.$$

Because $w^o \geq 0$, the stronger condition

$$p^N(A; F) \geq T$$

is sufficient for natural eligibility. It is not necessary: a firm can have $p^N < T$ and nevertheless naturally satisfy Factor R because positive owner wages are part of its unconstrained low-tax allocation.

D.2.1 Labor-intensity heterogeneity under standard technologies

It is useful to illustrate why the general specification $F_i(L, e)$ can generate substantial heterogeneity in natural employee-payroll intensity without imposing any particular degree of substitutability between inputs. The examples in this subsection are illustrative only and are not required for any proposition.

Consider a stripped-down version of the low-tax problem with no revenue underreporting, no owner wages, linear cost qe for the alternative input, and a locally constant favorable revenue-tax rate τ_L . Suppose the production function has decreasing returns,

with degree $\nu \in (0, 1)$. The firm's objective is

$$(1 - \tau_L)R(L, e) - wL - qe.$$

At an interior optimum,

$$w = (1 - \tau_L)R_L,$$

and therefore

$$\frac{wL}{R} = (1 - \tau_L) \frac{R_L L}{R}. \quad (52)$$

Natural employee-payroll intensity is therefore proportional to the output elasticity with respect to hired labor.

For example, consider a Cobb–Douglas technology

$$R = AL^{\nu\alpha}e^{\nu(1-\alpha)}.$$

Equation (52) gives

$$p^N = (1 - \tau_L)\nu\alpha. \quad (53)$$

Thus heterogeneity in the factor-distribution parameter α directly generates heterogeneity in natural employee-payroll intensity.

The same distinction between factor intensity and factor substitutability arises under CES production. Let

$$R = A[\alpha L^\rho + (1 - \alpha)e^\rho]^{\nu/\rho}, \quad \rho = \frac{\sigma - 1}{\sigma}, \quad \sigma > 0.$$

The labor output elasticity is

$$\frac{R_L L}{R} = \nu \frac{\alpha L^\rho}{\alpha L^\rho + (1 - \alpha)e^\rho}.$$

The interior first-order conditions imply

$$\frac{L}{e} = \left[\frac{\alpha q}{(1 - \alpha)w} \right]^\sigma,$$

so the odds of the labor component in the CES aggregator satisfy

$$\frac{s_L}{1 - s_L} = \left(\frac{\alpha}{1 - \alpha} \right)^\sigma \left(\frac{q}{w} \right)^{\sigma-1}, \quad s_L \equiv \frac{\alpha L^\rho}{\alpha L^\rho + (1 - \alpha)e^\rho}.$$

Hence s_L , and therefore $p^N = (1 - \tau_L)\nu s_L$, is strictly increasing in α for any $\sigma > 0$. Heterogeneity in factor intensity can therefore arise whether the two inputs are substitutes or

complements.

At the limiting case of perfect complements, consider

$$R = A \left[\min \left\{ \frac{L}{a_L}, \frac{e}{a_e} \right\} \right]^\nu.$$

Cost minimization implies $L = a_L z$ and $e = a_e z$. At the optimum,

$$\frac{wL}{R} = (1 - \tau_L) \nu \frac{wa_L}{wa_L + qa_e}.$$

Thus even with perfect complementarity, technologies can exhibit arbitrarily different labor intensities through their input requirements. These examples motivate treating labor intensity as an equilibrium feature of the general firm-specific technology F_i , rather than tying it to a particular elasticity of substitution.

D.3 Manufactured eligibility and scale

Consider now a firm whose unconstrained low-tax allocation does not naturally satisfy Factor R:

$$r^N(A; F) < T.$$

By Lemma D.2,

$$G(A; F) > 0 \iff B(A; F) > C(A; F).$$

Since $B > 0$, define

$$Q(A; F) = \frac{C(A; F)}{B(A; F)}. \quad (54)$$

Then

$$G(A; F) > 0 \iff Q(A; F) < 1. \quad (55)$$

D.3.1 Scale single crossing

Fix a non-scale technology F and consider a connected interval

$$\mathcal{I}(F) \subseteq \{A : r^N(A; F) < T\}.$$

Assumption 1 states that $Q(A; F)$ is continuous and strictly increasing in A on this interval and takes values both below and above one.

The restriction is imposed only within a connected non-natural-eligibility interval because the general framework does not impose monotonicity of $r^N(A; F)$ in A . As productivity changes, the firm's optimal production, employee payroll, owner compensation,

and reported revenue may all change. Consequently, the set

$$\{A : r^N(A; F) < T\}$$

need not itself be a single interval.

Lemma D.3 (Unique manufactured-eligibility cutoff). *Under Assumption 1, there exists a unique*

$$\bar{A}^M(F) \in \mathcal{I}(F)$$

such that

$$Q(\bar{A}^M(F); F) = 1.$$

Within $\mathcal{I}(F)$,

$$A < \bar{A}^M(F) \implies Q(A; F) < 1,$$

whereas

$$A > \bar{A}^M(F) \implies Q(A; F) > 1.$$

Proof. By assumption there exist $A_a, A_b \in \mathcal{I}(F)$ such that

$$Q(A_a; F) < 1 < Q(A_b; F).$$

Strict monotonicity implies $A_a < A_b$. Since $\mathcal{I}(F)$ is an interval and Q is continuous, the intermediate value theorem implies the existence of

$$\bar{A}^M(F) \in (A_a, A_b)$$

such that $Q(\bar{A}^M(F); F) = 1$. Strict monotonicity establishes uniqueness and the two inequalities on either side of the cutoff. $\square$

D.3.2 Differential interpretation of scale single crossing

Assumption 1 is a reduced-form restriction on optimized equilibrium objects. When B and C are differentiable and strictly positive,

$$\frac{\partial \log Q(A; F)}{\partial \log A} = \frac{\partial \log C(A; F)}{\partial \log A} - \frac{\partial \log B(A; F)}{\partial \log A}. \quad (56)$$

Thus a sufficient differential condition for Q to increase throughout an interval is

$$\varepsilon_{C,A} > \varepsilon_{B,A}. \quad (57)$$

This condition allows the firm's natural Factor R to change with scale. For example, if $r^N(A; F)$ rises with A , the proportional distance $T - r^N(A; F)$ may shrink as the firm becomes larger. Assumption 1 requires that, after accounting for this and all other endogenous responses, the optimized cost of satisfying Factor R rises relative to the value of the favorable tax schedule.

D.3.3 How primitives enter the scale condition

Although the sign of Q_A cannot be determined from the general technology alone, the derivatives of B and C show which primitives govern the scale comparison.

Let

$$z_j = R_j - \tilde{R}_j, \quad j \in \{H, L, T\}.$$

On differentiable branches, the envelope theorem gives

$$\frac{\partial \Pi_j(A; F)}{\partial A} = [1 - c'(z_j)] F(L_j, e_j), \quad j \in \{H, L, T\}. \quad (58)$$

For Π_T , the Factor R constraint does not depend directly on A , so its multiplier has no direct envelope term.

Define

$$M_j(A; F) \equiv [1 - c'(z_j)] F(L_j, e_j).$$

Then

$$B_A = M_L - M_H, \quad C_A = M_L - M_T. \quad (59)$$

Consequently,

$$Q_A > 0 \iff B C_A - C B_A > 0. \quad (60)$$

Equations (59)–(60) make clear that scale single crossing is a restriction on the full optimized problem rather than on any single primitive. The production technology F affects the allocations and their response to A . Revenue-reporting costs $c(\cdot)$ affect the marginal value of true revenue at each allocation. Labor costs $wL + \Phi(L)$ affect the cost of satisfying Factor R through employee payroll. Owner-compensation costs $\Gamma(w^o)$ affect the cost of satisfying it through owner wages. All these primitives also affect the natural Factor R $r^N(A; F)$ and therefore the distance from the unconstrained allocation to the eligibility threshold.

The owner-income-shifting case provides a transparent illustration. Suppose, only for this illustration, that the firm closes the eligibility gap by increasing owner wages while leaving the other components of its unconstrained low-tax allocation fixed. The required

additional owner wage is

$$m(A; F) = [T - r^N(A; F)] \tilde{R}^L(A; F). \quad (61)$$

Differentiating,

$$m_A = [T - r^N(A; F)] \tilde{R}_A^L - \tilde{R}^L r_A^N. \quad (62)$$

The first term is the scale effect: for a given proportional distance from the threshold, larger unconstrained revenue requires a larger nominal adjustment. The second captures endogenous movement in natural Factor R . If $r_A^N > 0$, this movement reduces the amount of additional owner wages required to qualify.

If the unconstrained owner wage is $w^{o,L}(A; F)$, the direct compensation cost of this illustrative adjustment is

$$\Gamma(w^{o,L} + m) - \Gamma(w^{o,L}).$$

Convexity of Γ increases the marginal cost of larger nominal adjustments. It can therefore generate a force toward C rising more rapidly with scale. It is not sufficient by itself for (57): production responses, changes in r^N , revenue reporting, labor adjustment, and the scaling of the tax benefit B can reinforce or offset this force. This is why Assumption 1 is stated in terms of the optimized ratio C/B in the main text.

D.4 Proof of Proposition 1

Proof. For natural eligibility, suppose

$$r_i^N \geq T.$$

The unconstrained low-tax allocation is feasible for the qualifying problem, so Lemma D.1 implies

$$\Pi_{T,i} = \Pi_{L,i}, \quad C_i = 0.$$

Since $B_i > 0$, Lemma D.2 gives

$$G_i = B_i > 0.$$

Moreover, because $\omega_i^N \geq 0$,

$$p_i^N \geq T \implies r_i^N = p_i^N + \omega_i^N \geq T,$$

so $p_i^N \geq T$ is sufficient for natural eligibility.

For manufactured eligibility, fix F_i and a connected interval of A over which

$$r_i^N(A) < T$$

and Assumption 1 holds. Lemma D.3 establishes a unique cutoff $\bar{A}^M(F_i)$ satisfying

$$Q_i(\bar{A}^M(F_i)) = 1.$$

For

$$A < \bar{A}^M(F_i),$$

strict monotonicity gives $Q_i(A) < 1$, or equivalently

$$C_i(A) < B_i(A).$$

Lemma D.2 therefore gives

$$G_i(A) = B_i(A) - C_i(A) > 0.$$

For

$$A > \bar{A}^M(F_i),$$

we instead have $Q_i(A) > 1$ and hence

$$C_i(A) > B_i(A),$$

so Lemma D.2 yields $G_i(A) = 0$.

At $A = \bar{A}^M(F_i)$,

$$B_i = C_i,$$

which implies

$$\Pi_{T,i} = \Pi_{H,i}.$$

The firm is therefore indifferent between satisfying Factor R and remaining under the high schedule. □

D.5 Incumbent firms and location around the threshold

Define the optimized low-tax payoff conditional on attaining Factor R r :

$$W_L(r) = \max_{x: r(x)=r} u_L(x). \tag{63}$$

Assumption 2 states that $W_L(r)$ is continuous and strictly single-peaked at the natural Factor R r^N .

Lemma D.4 (Best qualifying allocation). *Under Assumption 2,*

$$\arg \max_{r \geq T} W_L(r) = \begin{cases} r^N, & r^N \geq T, \\ T, & r^N < T. \end{cases}$$

Consequently,

$$\Pi_T = \begin{cases} \Pi_L, & r^N \geq T, \\ W_L(T), & r^N < T. \end{cases}$$

Proof. If $r^N \geq T$, the unconstrained maximizer of W_L belongs to the qualifying set and remains optimal after imposing $r \geq T$.

If $r^N < T$, every qualifying r lies to the right of the unique peak. Strict single-peakedness implies that $W_L(r)$ is strictly decreasing to the right of r^N . Hence the best qualifying value is attained at the smallest feasible Factor R , $r = T$. $\square$

The role of single-peakedness is therefore limited and precise. It is needed to conclude that a firm whose preferred low-tax Factor R lies below the threshold bunches exactly at T if it chooses to qualify. It is not needed for the value decomposition or for the no-downward-crossing result below.

D.6 Proof of Proposition 2

Let x^0 denote the selected pre-reform optimum under the high schedule and let

$$r^0 = r(x^0).$$

Because the pre-reform problem is the unrestricted high-schedule problem,

$$u_H(x^0) = \Pi_H.$$

Proof. First suppose $r^0 \geq T$. The pre-reform allocation x^0 is feasible for the post-reform qualifying problem. By schedule dominance,

$$u_L(x^0) > u_H(x^0) = \Pi_H.$$

Therefore

$$\Pi_T \geq u_L(x^0) > \Pi_H.$$

No allocation below the threshold can generate more than the unrestricted high-schedule maximum Π_H . Hence the firm's post-reform optimum cannot lie below T .

If $r^N \geq T$, Lemma D.4 implies that the best qualifying allocation is the unconstrained low-tax optimum, so

$$r^1 = r^N.$$

If instead $r^N < T$, the same lemma implies that the best qualifying allocation lies at the boundary,

$$r^1 = T.$$

Now suppose $r^0 < T$. If $r^N \geq T$, the unconstrained low-tax optimum is feasible under the eligibility requirement, and

$$\Pi_T = \Pi_L > \Pi_H.$$

The firm therefore qualifies and chooses

$$r^1 = r^N.$$

Finally, suppose both

$$r^0 < T \quad \text{and} \quad r^N < T.$$

By Lemma D.4, the best qualifying allocation has $r = T$ and value $\Pi_T = W_L(T)$. Since the pre-reform high-tax optimum itself satisfies $r^0 < T$, it remains feasible as a non-qualifying allocation after the reform and attains Π_H .

If $\Pi_T > \Pi_H$, equivalently $B > C$, the firm qualifies and bunches at

$$r^1 = T.$$

If $\Pi_T < \Pi_H$, equivalently $B < C$, it remains under the high schedule and retains its selected pre-reform optimum,

$$r^1 = r^0.$$

At equality, the firm is indifferent between the two alternatives. □

D.6.1 Why no downward crossing is a global result

The no-downward-crossing result does not rely on the firm being close to the threshold or on a local comparison around T . An initially above-threshold firm can retain its entire pre-reform allocation and receive the favorable schedule at that same allocation after the reform. Hence there exists a qualifying allocation whose payoff strictly exceeds the unrestricted maximum under the high schedule. This alone rules out any post-reform

optimum below T .

Single-peakedness is needed only to determine the location of the best qualifying allocation: r^N when $r^N \geq T$ and the boundary T when $r^N < T$.

D.7 Adjustment identity for threshold bunchers

Let

$$P = wL + w^o.$$

If a firm locates exactly at the Factor R threshold after the reform,

$$P^1 = T\tilde{R}^1.$$

Before the reform,

$$P^0 = r^0\tilde{R}^0.$$

Therefore

$$\begin{aligned}\Delta P &= T\tilde{R}^1 - r^0\tilde{R}^0 \\ &= T\Delta\tilde{R} + (T - r^0)\tilde{R}^0,\end{aligned}\tag{64}$$

which gives

$$\Delta P - T\Delta\tilde{R} = (T - r^0)\tilde{R}^0.\tag{65}$$

Holding the unit employee-payroll cost w fixed,

$$\Delta P = w\Delta L + \Delta w^o,$$

so

$$w\Delta L + \Delta w^o - T\Delta\tilde{R} = (T - r^0)\tilde{R}^0.\tag{66}$$

For a buncher coming from below, $r^0 < T$, so the right-hand side is positive. The firm must close its initial gap through some combination of higher employee payroll, higher owner wages, or lower reported revenue.

For a buncher coming from above, $r^0 > T$, so the right-hand side is negative. Such a firm reduces its payroll-to-revenue ratio relative to its initial allocation but is prevented by the eligibility requirement from moving below T .

Equation (66) is an accounting identity and does not determine which margin responds. The choice among margins is governed by the production, reporting, labor, and owner-compensation costs in the firm's optimization problem.

D.8 Selection into Simples

We next embed the operating problem into the firm's regime choice. Let K_S denote the fixed cost of operating under Simples. Before the reform,

$$V_S^0(A; F) = \Pi_H(A; F) - K_S. \quad (67)$$

By Lemma D.2, after the reform,

$$\begin{aligned} V_S^1(A; F) &= \max\{\Pi_H(A; F), \Pi_T(A; F)\} - K_S \\ &= V_S^0(A; F) + G(A; F). \end{aligned} \quad (68)$$

Thus the intensive and extensive margins are governed by exactly the same object: Factor R raises the value of selecting into Simples by $G(A; F)$.

D.9 Pre-reform productivity sorting

Let $V_P(A; F)$ denote the value of Lucro Presumido and normalize the outside option to zero. Define

$$D^0(A; F) \equiv V_P(A; F) - V_S^0(A; F). \quad (69)$$

Under Assumption 3, fixing F :

1. $V_S^0(A; F)$ is continuous and strictly increasing in A and crosses zero once at $A_E^0(F)$;
2. $D^0(A; F)$ is continuous and strictly increasing in A and crosses zero once at $A_{SP}^0(F)$;
3. $A_E^0(F) < A_{SP}^0(F)$.

Lemma D.5 (Pre-reform regime allocation). *Under Assumption 3, apart from exact indifference points,*

$$j^0(A; F) = \begin{cases} 0, & A < A_E^0(F), \\ S, & A_E^0(F) < A < A_{SP}^0(F), \\ P, & A > A_{SP}^0(F). \end{cases} \quad (70)$$

Proof. If $A < A_E^0(F)$, strict monotonicity implies

$$V_S^0(A; F) < 0.$$

Since $A < A_E^0(F) < A_{SP}^0(F)$, the unique-crossing property of D^0 also implies

$$D^0(A; F) < 0,$$

or $V_P < V_S^0 < 0$. The outside option therefore strictly dominates both formal regimes.

If

$$A_E^0(F) < A < A_{SP}^0(F),$$

then

$$V_S^0(A; F) > 0 \quad \text{and} \quad D^0(A; F) < 0.$$

Hence $V_S^0 > V_P$ and $V_S^0 > 0$, so Simple is optimal.

Finally, if $A > A_{SP}^0(F)$, then $D^0(A; F) > 0$ and therefore

$$V_P(A; F) > V_S^0(A; F).$$

Since $A > A_{SP}^0(F) > A_E^0(F)$, we also have $V_S^0(A; F) > 0$. Thus

$$V_P(A; F) > V_S^0(A; F) > 0,$$

and Lucro Presumido is optimal. □

D.10 Proof of Proposition 3

A firm is induced into Simple if its post-reform regime is S but its pre-reform regime is either the outside option or Lucro Presumido.

Proof. Fix the firm's non-scale technology F_i .

If an induced firm's pre-reform counterfactual is the outside option, Lemma D.5 implies

$$A_i < A_E^0(F_i).$$

If instead its pre-reform counterfactual is Lucro Presumido, the same lemma implies

$$A_i > A_{SP}^0(F_i).$$

Since

$$A_E^0(F_i) < A_{SP}^0(F_i),$$

an outside-option switcher i and a Lucro-Presumido switcher k with the same non-scale technology satisfy

$$A_i < A_E^0(F_i) < A_{SP}^0(F_i) < A_k.$$

Hence, conditional on the same non-scale technology, firms induced from Lucro Presumido have higher potential scale than firms induced from the outside option. □

D.11 Movement of the regime boundaries

Proposition 3 requires only pre-reform sorting. Additional conditions are needed to represent the reform as movements in unique post-reform regime boundaries.

Define

$$D^1(A; F) \equiv V_P(A; F) - V_S^1(A; F) = D^0(A; F) - G(A; F).$$

Suppose $V_S^1(A; F)$ is locally continuous and strictly increasing in A and has a unique zero $A_E^1(F)$. At the old outside–Simples boundary,

$$V_S^1(A_E^0(F); F) = V_S^0(A_E^0(F); F) + G(A_E^0(F); F) = G(A_E^0(F); F).$$

Therefore, if

$$G(A_E^0(F); F) > 0, \tag{71}$$

then

$$V_S^1(A_E^0(F); F) > 0.$$

Since V_S^1 is increasing and equals zero at $A_E^1(F)$,

$$A_E^1(F) < A_E^0(F). \tag{72}$$

Similarly, suppose $D^1(A; F)$ is locally continuous and strictly increasing and has a unique zero $A_{SP}^1(F)$. At the old Simples–Lucro Presumido boundary,

$$D^1(A_{SP}^0(F); F) = -G(A_{SP}^0(F); F).$$

Thus, if

$$G(A_{SP}^0(F); F) > 0, \tag{73}$$

then

$$D^1(A_{SP}^0(F); F) < 0,$$

and strict monotonicity implies

$$A_{SP}^1(F) > A_{SP}^0(F). \tag{74}$$

Factor R therefore expands the interval over which Simples is preferred whenever it raises the value of Simples at the relevant old boundary and the post-reform single-crossing structure is preserved. The additional post-reform conditions are needed only to describe the response in terms of unique boundary movements; they are not required for Proposition 3.

D.12 Natural and manufactured eligibility across entry margins

Propositions 1 and 3 characterize distinct dimensions of selection. The former describes the mechanism through which Factor R raises the value of Simples; the latter describes the regime a marginal firm would have chosen in the absence of the reform.

There is no general one-to-one mapping between these dimensions. A firm induced from either the outside option or Lucro Presumido may be naturally eligible or may find it profitable to manufacture eligibility. Both the regime boundaries and the manufactured-eligibility cutoff depend on the firm's non-scale technology F .

The interaction can nevertheless be characterized conditional on a given F . Consider a non-naturally eligible firm near the old Simples–Lucro Presumido boundary. To be attracted to Simples specifically through manufactured eligibility it must satisfy

$$r^N(A; F) < T \quad \text{and} \quad Q(A; F) < 1. \quad (75)$$

Within a connected non-natural-eligibility interval satisfying Assumption 1,

$$Q(A; F) < 1 \iff A < \bar{A}^M(F).$$

A Lucro-Presumido switcher must simultaneously satisfy

$$A > A_{SP}^0(F).$$

Hence manufactured eligibility can induce firms from Lucro Presumido only if the two regions overlap:

$$A_{SP}^0(F) < \bar{A}^M(F), \quad (76)$$

with the relevant values of A also lying inside the non-natural-eligibility interval.

Condition (76) is not imposed by the general framework. Nor does the framework restrict how frequently it holds across different technologies F . Similarly, firms near the upper regime boundary may instead be naturally eligible if their technology generates $r^N \geq T$. Which mechanism dominates among firms induced from each counterfactual regime is therefore an empirical question.

D.13 Role of the assumptions

For completeness, we summarize the role of each restriction used in the main text.

Schedule dominance. The condition

$$\mathcal{T}_L(\tilde{R}) < \mathcal{T}_H(\tilde{R})$$

establishes $B > 0$. Together with the feasibility of an initially above-threshold firm's old allocation, it is also sufficient to rule out downward crossing by such incumbent firms.

Scale single crossing. Assumption 1 requires

$$Q(A; F) = \frac{C(A; F)}{B(A; F)}$$

to increase with A within a connected region of firms that do not naturally qualify, holding the non-scale technology F fixed. It generates the manufactured-eligibility cutoff $\bar{A}^M(F)$. No monotonicity of natural Factor R in A is imposed.

Factor R single-peakedness. Assumption 2 determines the location of a constrained qualifying firm's optimum. If $r^N < T$, it implies that the best qualifying allocation lies exactly at $r = T$. It is not required for the value decomposition $G = [B - C]_+$ or for the global no-downward-crossing result.

Productivity sorting across regimes. Assumption 3 orders the firm's pre-reform alternatives, conditional on its non-scale technology, as

$$0 \longrightarrow S \longrightarrow P$$

as A increases. Proposition 3 follows from this pre-reform ordering. No restriction on the shape of $G(A; F)$ is required for the proposition.

Additional post-reform single-crossing conditions are needed only if the reform is represented as movements in unique regime boundaries.

E Decomposition for new entrants

Assuming standard monotonicity (the policy never deters entry of a firm that would otherwise have entered), the potential entrants partition into three types:

$$\begin{aligned} \text{never-takers (NT): } & D(0) = 0, D(1) = 0, \\ \text{compliers (C): } & D(0) = 0, D(1) = 1, \\ \text{always-takers (AT): } & D(0) = 1, D(1) = 1. \end{aligned}$$

Where $D_{ic}(w) \in \{0, 1\}$ is the entry indicator of potential firm i under treatment status w . The compliers are the *marginal entrants* (they enter because of the policy); the always-takers are the *inframarginal entrants* (they would have entered in any case). Because the policy may also have changed how the inframarginal entrants behave once they entered, separately identifying the average treatment effect on the always-takers (to the entry decision) from the compliers' characteristics is challenging.

In a treated industry after 2018 the set of *observed* new firms is $AT \cup C$, and for each we observe any outcome $Y(1)$, such as the Factor R value. Absent the policy, the same industry's new firms would have been the always-takers only, with outcome $Y(0)$. Define

$$M \equiv \mathbb{E}[Y(1) \mid AT \cup C], \quad \bar{Y}_{AT}^0 \equiv \mathbb{E}[Y(0) \mid AT].$$

M is the mean outcome of treated post-period entrants, read directly from the data. $\bar{Y}_{AT}^0$ is the mean outcome the treated industries' entrants would have had absent the policy. The post-period event-study coefficient β_k from the outcome regression identifies

$$\beta_k^{DiD} = M_k - \bar{Y}_{AT}^0.$$

Let N_1 be the number of new firms in a treated industry-year after 2018 and let N_0 be the counterfactual count absent the policy, i.e. the number of always-takers. By definition $N_1 = N_{AT} + N_C$ and $N_0 = N_{AT}$, so the count difference-in-differences identifies the number of marginal entrants directly:

$$N_C = N_1 - N_0,$$

where $N_1 - N_0$ is exactly the post-period coefficient of the count regression. The type shares among observed treated entrants are therefore *point-identified*:

$$\pi_{AT} = \frac{N_{AT}}{N_1} = \frac{N_0}{N_1}, \quad \pi_C = \frac{N_C}{N_1} = \frac{N_1 - N_0}{N_1}, \quad \pi_{AT} + \pi_C = 1.$$

The observed treated mean is the type-weighted average of the two groups' treated outcomes:

$$M = \pi_{AT} \mathbb{E}[Y(1) \mid AT] + \pi_C \mathbb{E}[Y(1) \mid C]. \quad (77)$$

Subtracting $\bar{Y}_{AT}^0 = (\pi_{AT} + \pi_C) \bar{Y}_{AT}^0$ from both sides and adding and subtracting $\pi_C \mathbb{E}[Y(1) \mid AT]$ gives the target decomposition of the outcome DiD:

$$\beta^{DiD} = \underbrace{\left(\mathbb{E}[Y(1) \mid AT] - \bar{Y}_{AT}^0 \right)}_{(I)} + \underbrace{\pi_C \left(\mathbb{E}[Y(1) \mid C] - \mathbb{E}[Y(1) \mid AT] \right)}_{(II)}. \quad (78)$$

Term (I) is the intensive-margin response of firms that would have entered regardless.

We denote this behavioral response by $\Delta_{AT} \equiv E[Y(1) | AT] - E[Y(0) | AT]$. Term (II) is the contribution of composition to the reduced-form effect. Dividing it by the complier share delivers our main object of interest: the selection gap. It reveals how those new entrants induced by the policy differ from those firm that would have entered anyway. Their differences in behavior and characteristics shed light on the entry motives.

To isolate the two terms in Equation 78, we rely on two additional pieces of information. First, for outcomes that may respond to the policy, we use the behavioral responses of incumbent firms estimated in Section 5. Rather than assuming that inframarginal entrants would not have changed their behavior after the reform, we assume that, absent the reform, the Low- versus High-FR composition of inframarginal entrants would match that observed among pre-reform entrants, and that conditional on this status their behavioral response equals the corresponding incumbent response. In particular, we use the estimated incumbent responses in reported revenues, employees' payroll, and owner wages to recover term (I) in Equation 78.⁴ For outcomes that are predetermined with respect to the policy, we instead set $\Delta_{AT} = 0$.

Second, the difference-in-differences estimate for the number of new firms identifies the share of observed post-reform entrants that are marginal to the policy. Let N_1 denote the number of entrants observed under the reform and N_0 the number that would have entered absent the reform. Since

$$N_1 = N_{AT} + N_C \quad \text{and} \quad N_0 = N_{AT},$$

the complier share among observed entrants is

$$\pi_C = \frac{N_C}{N_1} = \frac{N_1 - N_0}{N_1}. \quad (79)$$

In our empirical implementation, using the log-number-of-firms specification,

$$\pi_C = 1 - \exp\{-\beta^{\log N}\}.$$

Given Δ_{AT} and π_C , Equation (78) directly identifies term (II):

$$(II) = \beta^{DiD} - \Delta_{AT} = \pi_C [E[Y(1) | C] - E[Y(1) | AT]]. \quad (80)$$

Dividing by the share of marginal entrants therefore recovers the difference between marginal

⁴This is not a theoretical assumption, but rather an empirical assumption. We do not observe how inframarginal new firms would have behaved in the absence of the policy, but we rely on our incumbents' responses as a proxy.

and inframarginal entrants under the same post-reform environment:

$$E[Y(1) | C] - E[Y(1) | AT] = \frac{\beta^{DiD} - \Delta_{AT}}{\pi_C}. \quad (81)$$

We refer to this object as the *selection gap*. It is our main object for characterizing entry motives: it measures whether the firms induced to enter by Factor R differ from firms that would have entered Simples regardless of the reform, after netting out the behavioral response to the policy that both groups may exhibit.

A second useful comparison asks how marginal entrants under the reform differ from the entrants that would have been observed in its absence. Combining the selection gap with the behavioral response of inframarginal entrants gives

$$E[Y(1) | C] - E[Y(0) | AT] = \Delta_{AT} + \frac{\beta^{DiD} - \Delta_{AT}}{\pi_C}. \quad (82)$$

The selection gap therefore isolates differences in composition under the same policy environment, while Equation (82) compares the firms induced to enter by Factor R with the entrants that would have been observed in its absence.

F Additional Data and Context Details

e-Social:

It is a unified, electronic reporting system for employers that collects high-frequency information on formal labor relationships—hires and separations, monthly earnings, hours, and other payroll-relevant events—together with the information needed to assess payroll-based contributions and labor obligations. Conceptually, it is the “digital backbone” of employer compliance: instead of filling multiple separate payroll forms, firms transmit standardized event-level and monthly payroll records through a single platform, which are then shared across the relevant government administrations (tax collection, social security, and labor authorities). In practice, eSocial replaced the older payroll filing (GFIP) for many taxpayers and functions, shifting payroll compliance from a standalone monthly form toward an integrated, electronic reporting pipeline that supports both calculation and enforcement.

DECRED:

It is a third-party information return on card payments, filed by card networks/acquirers (the payment processors), not by firms, and submitted to the federal tax authority. It

reports, for each merchant (identified by tax ID), the aggregate amounts processed via payment cards by month, allowing the tax authority to cross-check firms' self-reported sales against independently reported electronic payments.

G Additional Tables

Table G.6: Simplified summary of the 2018 Simples Nacional reform

Pre-Reform (Old) Anexo	Post-Reform Outcome
Anexo I	Largely unchanged; only very small adjustments in the tax schedule.
Anexo II	Largely unchanged; only very small adjustments in the tax schedule.
Anexo III	Following the reform, activities previously classified under Old Anexo III are split into two groups. Activities classified as <i>not subject to the Fator R</i> remain in New Anexo III. Activities classified as <i>subject to the Fator R</i> remain in New Anexo III only if their payroll-to-revenue ratio exceeds the Fator R threshold; otherwise, they are reclassified into New Anexo V.
Anexo IV	Largely unchanged; only very small adjustments in the tax schedule.
Anexo V	Following the reform, activities previously classified under Old Anexo V are also split into two groups. Activities classified as <i>subject to the Fator R</i> migrate to New Anexo III if their payroll-to-revenue ratio exceeds the Fator R threshold; otherwise, they remain in New Anexo V. In addition, some activities are reclassified by statutory listing and may move directly to New Anexo III or New Anexo IV, independently of the Fator R.
Anexo VI	Firms are mechanically reassigned to New Anexo III or V based on the Fator R following the extinction of New Anexo VI.

Table G.7: Anexo I: Commerce

	Receita Bruta em 12 Meses (em R\$)	Alíquota	Valor a Deduzir (em R\$)
1ª Faixa	Até 180.000,00	4,00%	-
2ª Faixa	De 180.000,01 a 360.000,00	7,30%	5.940,00
3ª Faixa	De 360.000,01 a 720.000,00	9,50%	13.860,00
4ª Faixa	De 720.000,01 a 1.800.000,00	10,70%	22.500,00
5ª Faixa	De 1.800.000,01 a 3.600.000,00	14,30%	87.300,00
6ª Faixa	De 3.600.000,01 a 4.800.000,00	19,00%	378.000,00

Table G.8: Anexo II: Manufacturing

	Receita Bruta em 12 Meses (em R\$)	Alíquota	Valor a Deduzir (em R\$)
1ª Faixa	Até 180.000,00	4,50%	-
2ª Faixa	De 180.000,01 a 360.000,00	7,80%	5.940,00
3ª Faixa	De 360.000,01 a 720.000,00	10,00%	13.860,00
4ª Faixa	De 720.000,01 a 1.800.000,00	11,20%	22.500,00
5ª Faixa	De 1.800.000,01 a 3.600.000,00	14,70%	85.500,00
6ª Faixa	De 3.600.000,01 a 4.800.000,00	30,00%	720.000,00

Table G.9: Anexo III: serviços não relacionados no 5o-C do art. 18 desta Lei Complementar

	Receita Bruta em 12 Meses (em R\$)	Alíquota	Valor a Deduzir (em R\$)
1ª Faixa	Até 180.000,00	6,00%	–
2ª Faixa	De 180.000,01 a 360.000,00	11,20%	9.360,00
3ª Faixa	De 360.000,01 a 720.000,00	13,50%	17.640,00
4ª Faixa	De 720.000,01 a 1.800.000,00	16,00%	35.640,00
5ª Faixa	De 1.800.000,01 a 3.600.000,00	21,00%	125.640,00
6ª Faixa	De 3.600.000,01 a 4.800.000,00	33,00%	648.000,00

Table G.10: Anexo IV: serviços relacionados no 5o-C do art. 18 desta Lei Complementar

	Receita Bruta em 12 Meses (em R\$)	Alíquota	Valor a Deduzir (em R\$)
1ª Faixa	Até 180.000,00	4,50%	–
2ª Faixa	De 180.000,01 a 360.000,00	9,00%	8.100,00
3ª Faixa	De 360.000,01 a 720.000,00	10,20%	12.420,00
4ª Faixa	De 720.000,01 a 1.800.000,00	14,00%	39.780,00
5ª Faixa	De 1.800.000,01 a 3.600.000,00	22,00%	183.780,00
6ª Faixa	De 3.600.000,01 a 4.800.000,00	33,00%	828.000,00

Table G.11: Anexo V: serviços relacionados no 5o-I do art. 18 desta Lei Complementar

	Receita Bruta em 12 Meses (em R\$)	Alíquota	Valor a Deduzir (em R\$)
1ª Faixa	Até 180.000,00	15,50%	–
2ª Faixa	De 180.000,01 a 360.000,00	18,00%	4.500,00
3ª Faixa	De 360.000,01 a 720.000,00	19,50%	9.900,00
4ª Faixa	De 720.000,01 a 1.800.000,00	20,50%	17.100,00
5ª Faixa	De 1.800.000,01 a 3.600.000,00	23,00%	62.100,00
6ª Faixa	De 3.600.000,01 a 4.800.000,00	30,50%	540.000,00

Table G.12: Anexo VI: prestação de serviços relacionados no 5o-I do art. 18 desta Lei Complementar

Receita Bruta em 12 meses (em R\$)	Alíquota	IRPJ, PIS/Pasep, CSLL, Cofins e CPP	ISS
Até 180.000,00	16,93%	14,93%	2,00%
De 180.000,01 a 360.000,00	17,72%	14,93%	2,79%
De 360.000,01 a 540.000,00	18,43%	14,93%	3,50%
De 540.000,01 a 720.000,00	18,77%	14,93%	3,84%
De 720.000,01 a 900.000,00	19,04%	15,17%	3,87%
De 900.000,01 a 1.080.000,00	19,94%	15,71%	4,23%
De 1.080.000,01 a 1.260.000,00	20,34%	16,08%	4,26%
De 1.260.000,01 a 1.440.000,00	20,66%	16,35%	4,31%
De 1.440.000,01 a 1.620.000,00	21,17%	16,56%	4,61%
De 1.620.000,01 a 1.800.000,00	21,38%	16,73%	4,65%
De 1.800.000,01 a 1.980.000,00	21,86%	16,86%	5,00%
De 1.980.000,01 a 2.160.000,00	21,97%	16,97%	5,00%
De 2.160.000,01 a 2.340.000,00	22,06%	17,06%	5,00%
De 2.340.000,01 a 2.520.000,00	22,14%	17,14%	5,00%
De 2.520.000,01 a 2.700.000,00	22,21%	17,21%	5,00%
De 2.700.000,01 a 2.880.000,00	22,21%	17,21%	5,00%
De 2.880.000,01 a 3.060.000,00	22,32%	17,32%	5,00%
De 3.060.000,01 a 3.240.000,00	22,37%	17,37%	5,00%
De 3.240.000,01 a 3.420.000,00	22,41%	17,41%	5,00%
De 3.420.000,01 a 3.600.000,00	22,45%	17,45%	5,00%

Table G.13: Anexo I pre-reform

Até 180.000,00	4,00%
De 180.000,01 a 360.000,00	5,47%
De 360.000,01 a 540.000,00	6,84%
De 540.000,01 a 720.000,00	7,54%
De 720.000,01 a 900.000,00	7,60%
De 900.000,01 a 1.080.000,00	8,28%
De 1.080.000,01 a 1.260.000,00	8,36%
De 1.260.000,01 a 1.440.000,00	8,45%
De 1.440.000,01 a 1.620.000,00	9,03%
De 1.620.000,01 a 1.800.000,00	9,12%
De 1.800.000,01 a 1.980.000,00	9,95%
De 1.980.000,01 a 2.160.000,00	10,04%
De 2.160.000,01 a 2.340.000,00	10,13%
De 2.340.000,01 a 2.520.000,00	10,23%
De 2.520.000,01 a 2.700.000,00	10,32%
De 2.700.000,01 a 2.880.000,00	11,23%
De 2.880.000,01 a 3.060.000,00	11,32%
De 3.060.000,01 a 3.240.000,00	11,42%
De 3.240.000,01 a 3.420.000,00	11,51%
De 3.420.000,01 a 3.600.000,00	11,61%

Table G.14: Anexo II pre-reform

Até 180.000,00	4,50%
De 180.000,01 a 360.000,00	5,97%
De 360.000,01 a 540.000,00	7,34%
De 540.000,01 a 720.000,00	8,04%
De 720.000,01 a 900.000,00	8,10%
De 900.000,01 a 1.080.000,00	8,78%
De 1.080.000,01 a 1.260.000,00	8,86%
De 1.260.000,01 a 1.440.000,00	8,95%
De 1.440.000,01 a 1.620.000,00	9,53%
De 1.620.000,01 a 1.800.000,00	9,62%
De 1.800.000,01 a 1.980.000,00	10,45%
De 1.980.000,01 a 2.160.000,00	10,54%
De 2.160.000,01 a 2.340.000,00	10,63%
De 2.340.000,01 a 2.520.000,00	10,73%
De 2.520.000,01 a 2.700.000,00	10,82%
De 2.700.000,01 a 2.880.000,00	11,73%
De 2.880.000,01 a 3.060.000,00	11,82%
De 3.060.000,01 a 3.240.000,00	11,92%
De 3.240.000,01 a 3.420.000,00	12,01%
De 3.420.000,01 a 3.600.000,00	12,11%

Table G.15: Anexo III pre-reform (didn't include 5o-D)

Receita Bruta em 12 meses (em R\$)	ALÍQUOTA
Até 180.000,00	6,00%
De 180.000,01 a 360.000,00	8,21%
De 360.000,01 a 540.000,00	10,26%
De 540.000,01 a 720.000,00	11,31%
De 720.000,01 a 900.000,00	11,40%
De 900.000,01 a 1.080.000,00	12,42%
De 1.080.000,01 a 1.260.000,00	12,54%
De 1.260.000,01 a 1.440.000,00	12,68%
De 1.440.000,01 a 1.620.000,00	13,55%
De 1.620.000,01 a 1.800.000,00	13,68%
De 1.800.000,01 a 1.980.000,00	14,93%
De 1.980.000,01 a 2.160.000,00	15,06%
De 2.160.000,01 a 2.340.000,00	15,20%
De 2.340.000,01 a 2.520.000,00	15,35%
De 2.520.000,01 a 2.700.000,00	15,48%
De 2.700.000,01 a 2.880.000,00	16,85%
De 2.880.000,01 a 3.060.000,00	16,98%
De 3.060.000,01 a 3.240.000,00	17,13%
De 3.240.000,01 a 3.420.000,00	17,27%
De 3.420.000,01 a 3.600.000,00	17,42%

Table G.16: Anexo IV pre-reform

Receita Bruta em 12 meses (em R\$)	Alíquota
Até 180.000,00	4,50%
De 180.000,01 a 360.000,00	6,54%
De 360.000,01 a 540.000,00	7,70%
De 540.000,01 a 720.000,00	8,49%
De 720.000,01 a 900.000,00	8,97%
De 900.000,01 a 1.080.000,00	9,78%
De 1.080.000,01 a 1.260.000,00	10,26%
De 1.260.000,01 a 1.440.000,00	10,76%
De 1.440.000,01 a 1.620.000,00	11,51%
De 1.620.000,01 a 1.800.000,00	12,00%
De 1.800.000,01 a 1.980.000,00	12,80%
De 1.980.000,01 a 2.160.000,00	13,25%
De 2.160.000,01 a 2.340.000,00	13,70%
De 2.340.000,01 a 2.520.000,00	14,15%
De 2.520.000,01 a 2.700.000,00	14,60%
De 2.700.000,01 a 2.880.000,00	15,05%
De 2.880.000,01 a 3.060.000,00	15,50%
De 3.060.000,01 a 3.240.000,00	15,95%
De 3.240.000,01 a 3.420.000,00	16,40%
De 3.420.000,01 a 3.600.000,00	16,85%

Table G.17: Anexo V Pre-reform

RB em 12 meses	(r) < 0,10	0,10 – 0,15	0,15 – 0,20	0,20 – 0,25	0,25 – 0,30	0,30 – 0,35	0,35 – 0,40	(r) ≥ 0,40
Até 180k	17,50%	15,70%	13,70%	11,82%	10,47%	9,97%	9,10%	8,00%
180k a 360k	17,52%	15,75%	13,90%	12,60%	12,33%	10,72%	9,10%	8,48%
360k a 540k	17,55%	15,95%	14,20%	12,90%	12,64%	11,14%	9,58%	9,03%
540k a 720k	17,95%	16,70%	15,00%	13,70%	13,45%	12,00%	10,56%	9,34%
720k a 900k	18,15%	16,95%	15,30%	14,03%	13,53%	12,40%	11,04%	10,06%
900k a 1.080k	18,45%	17,20%	15,40%	14,10%	13,60%	12,60%	11,60%	10,60%
1.080k a 1.260k	18,55%	17,30%	15,50%	14,11%	13,68%	12,69%	11,69%	10,69%
1.260k a 1.440k	18,62%	17,32%	15,60%	14,12%	13,69%	12,69%	11,69%	10,69%
1.440k a 1.620k	18,72%	17,42%	15,70%	14,13%	14,08%	13,09%	12,08%	11,08%
1.620k a 1.800k	18,86%	17,56%	15,80%	14,14%	14,09%	13,09%	12,09%	11,09%
1.800k a 1.980k	18,96%	17,66%	15,90%	14,49%	14,45%	13,61%	12,78%	11,87%
1.980k a 2.160k	19,06%	17,76%	16,00%	14,67%	14,64%	13,79%	13,15%	12,28%
2.160k a 2.340k	19,26%	17,96%	16,20%	14,86%	14,82%	14,17%	13,54%	12,68%
2.340k a 2.520k	19,56%	18,30%	16,50%	15,46%	15,18%	14,61%	14,04%	13,26%
2.520k a 2.700k	20,79%	19,30%	17,45%	16,24%	16,00%	15,52%	15,03%	14,29%
2.700k a 2.880k	21,20%	20,00%	18,28%	16,91%	16,72%	16,32%	15,93%	15,23%
2.880k a 3.060k	21,70%	20,50%	18,70%	17,40%	17,13%	16,82%	16,38%	16,17%
3.060k a 3.240k	22,20%	21,00%	19,10%	17,90%	17,55%	17,22%	16,82%	16,51%
3.240k a 3.420k	22,50%	21,30%	19,50%	18,20%	17,97%	17,44%	17,24%	16,94%
3.420k a 3.600k	22,90%	21,80%	20,00%	18,60%	18,40%	17,85%	17,60%	17,18%